

Deep learning framework for predictive modeling of crude oil price for sustainable management in oil markets

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ABSTRACT

Crude oil price predictability has continually been considered as a fundamental argument of finance literature, given its critical propositions for risk management, investment decisions, and commercial and financial policymaking. This work presents an innovative learning framework for efficient predictive modeling of daily and weekly crude oil price (COP) information, which aims to enable sustainable management in oil markets. Firstly, an optimized version of variation mode decomposition (OVMD) is proposed to adaptively decompose the original COP time series into multiple modes based on a set of optimized parameters calculated with a Tree-structured Parzen Estimator (TPE) algorithm. Secondly, an AdaBoost algorithm is redesigned using random forest (RF) to model the future price information in the modes with the high frequency. Thirdly, a new deep network is presented to develop automatically learn spatial-temporal representations from decomposed COP data, where a novel Conv-former module is designed to efficiently extract local as well as global spatial representations without incurring extra computational costs. Following, Multiple Long short-term Memory (LSTM) networks are stacked to learn temporal representations from input modes. To further empower the representation power of our framework, a new bidirectional learning module is presented to stack the LSTM layer to learn from COP data in the forward and backward directions. To validate the efficiency of the proposed framework, this work performs experimental simulations and analyses based on a case study from Brent crude oil prices at both daily and weekly scales. The experimental findings show up the competent predictive modeling capabilities of the proposed framework over the cutting-edge methods rendering it as a promising solution to enable sustainable management in crude oil markets. The proposed framework can be generalized to different predictive modeling tasks and hence qualified to be used as a valuable tool for oil portfolio creation, property pricing, and risk management in Crude Oil Markets.

1. Introduction

Crude oil is a crucial source of energy for trade and industry development. The violent fluctuations in the crude oil price (COP) have considerable influences on the economic steadiness and even public security of nations, leading to substantial challenges related to international financial constancy and industrial management (Cai, Hamori, Yang, & Tian, 2020; Huynh, Shahbaz, Nasir, & Ullah, 2020). As an “industrial blood”, the effect of COP oscillations is considerable. For instance, the radical variations in international COP have completely influenced the sustainability of social development and economic growth (Chai, Wang, Wang, & Wang, 2019). The research efforts and economic community demonstrated that under the oil price transmission process, COP volatility presumes substantial impacts on the economic growth rate, joblessness ratio, monetary policy, and cost

value. To achieve the viable development of financial trading, economy, and risk management, the development behavior of future oil prices is indispensable information that must be considered in the task of designating sustainable development policy. The finance minister of Nigeria grieved that it is the inadequate predictive modeling of COP price and overreckoning of oil production is the main reason behind for imperfect management of yearly budgets in Nigeria (Ayadi, Williams, & Hyman, 2009). Thus, the objective of this study is to design an innovative high-precision predictive modeling approach to offer the information of COP trends for the government to devise sustainable development policy.

Developing a robust predictive modeling approach for COP is a promising solution to deal with these challenges. It may seem simple for the community to predict the COP. However, it persists as a challenging task because of its inherent uncertainty, nonlinearity, and chaos

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characteristic caused by different considerations, involving the need and supply state, geopolitical tensions (Jue Wang, Li, Hong, & Wang, 2018), hypothetical trading (Sun, Sun, Wang, & Wei, 2018), risk factors, and economic development (Zhao, Li, & Yu, 2017). Besides, uncertainty and instability of COP can also stem from pandemic situations (i.e., COVID-19), which might influence the investors' decisions for portfolio distribution and manufacturers' decisions for economy and business production. An operative uncertainty modeling of the COP time-series is beneficial to the technical development of the crude oil market and decrease of charges concerning decision-making risks, and therefore it gains cumulative interest in both academia and industry.

Though the complicated above-mentioned considerations make COP prediction a challenging task, the related research studies have been revealing many achievements. This work taxonomizes the research studies related to predictive modeling of COP into four categories methods, specifically, conventional econometric methods, machine learning (ML) methods, deep learning (DL) methods, and hybrid methods. The econometric methods can be described by regression principles and statistical distribution philosophy. typical examples of econometric methods include autoregressive integrated moving average model (ARIMA) (Moshiri & Foroutan, 2006), error correction model (Carpio, 2019), generalized autoregressive conditional heteroskedasticity (GARCH) (Abdollahi, 2020), Autoregressive Fractionally Integrated Moving Average (ARFIMA) (Pumi, Valk, Bisognin, Bayer, & Prass, 2019), etc. Though these econometric methods show acceptable computational performance, the linear type and limited distribution hypothesis are complicated to deal with irregular time-series, (i.e., COP), making them unable to achieve ideal predictive performance. In addition, the further development of predictive capability utilizing econometric methods encounters technical complexities. For instance, they are unable to be adaptively learned and just re-determine their parameters upon the underlying dataset change. The econometric models successfully concentrate on the linearity and volatility of COP data (Abdollahi, 2020) but are unable to take over the non-stationarity and nonlinearity present in real-world crude oil markets (Huynh et al., 2020; J. Li & Wang, 2020). Thus, all these factors make the financial and management stakeholders cannot trust and depend on the decisions made by these statistical methods limiting their applicability in crude oil markets.

As a reaction proceeds, the research community proposes using data-driven ML solutions to cope with the technical characteristics of the COP time series (Moosavi, Khasteh, & Bagheri, 2015). This mainly concentrates on using ML algorithms to handle complicated predictions by permitting machines to imitate human intelligence or brain learning methodology to gain more efficient and precise findings. Therefore, ML has been witnessing a wide acceptance as in developing a solution for predictive modeling of international COP using a variety of ML algorithms have been proposed including artificial neural networks (ANNs), Support vector machine (SVR), adaptive neuro-fuzzy inference system (ANFIS), extreme learning machine (ELM), AdaBoost. Although numerous studies have demonstrated that ML algorithms can efficiently analyze complicated systems, they also exhibit many shortcomings, such as over-fitting, local optimization, parameter sensitivity, generalizability, the curse of dimensionality, and sensitivity to adopted feature engineering.

Deep learning (DL) provides an excellent remedy for the above problems owing to its high analytical power and outstanding capability to model complicated nonlinear time-series. Convolutional neural networks (CNNs), recurrent neural networks (RNNs) are prevailing DL models that demonstrate great achievements in financial research. Among RNNs, Longest Short-term Memory (LSTM) and gated recurrent units (GRU) are broadly studied for predictive modeling from financial data (Rezaei, Faaljou, & Mansourfar, 2021). Nevertheless, DL has not obtained sufficient research attention for the deterministic and probabilistic prediction of COP, which may obstruct further improvement in the area. Motivated by that, this study proposes to take the advantage of

recent advances in DL to obtain reliable and accurate forecasting of COP that can empower financial decision-making and risk management.

1.1. Research gaps

- Because of the nonlinear, dynamic, and erratic characteristics of dynamic crude oil markets, determining its fundamental cyclic patterns of oil prices can be difficult. In the realm of market change, predictive analysis, which could also expose its underlying price trends from both descriptive and analytical viewpoints, has not gained nearly enough attention as it should have.
- Detecting anomalies and outliers in nonlinear COP time series is important for increasing the prediction model's generalization and efficacy. Discovering anomalies for COP time series is required since the inconsistencies generated by many components in COP frequently lead to considerable changes in the COP, increasing the chances of anomalies appearing. However, detecting anomalies for the COP has not been a primary focus of recent study in the subject.
- Because of the complicated characteristics of the COP, feature selection for doing predictive modeling of the COP data has become restricted. For many literature studies, the partial autocorrelation function (PACF) was used to find the most acceptable input features to use for predictive modeling. The basic computing process of the PACF, on the other hand, inhibits its capacity to deal with complicated dynamic COP time series.
- Predicting modeling of financial time series in oil markets is a problematic task because of the long-duration recall, dynamicity, and non-linearity, which restrict the predictive power of conventional econometric models and complicate the associated managerial tasks and decisions.

1.2. Aim and contributions

This research seeks to take the advantage of recent advances in deep learning to design an innovative solution for predictive modeling in crude oil markets. The main contributions of this work can be summarized as follow:

- This study proposes a new learning framework that can effectively improve the predictive modeling of dynamic, non-linear, and composite characteristics of COP time series. The design of this framework aims to offer the community an efficient tool for generating insightful information about the future of COP, which can help automate the decision-making, risk analysis, and other managerial tasks in oil markets at both local and global scales.
- This work introduces a novel decomposition method, called OVMD, which extends the standard variational mode decomposition, by adaptively and automatically selecting the corresponding parameters in such a way that improves predictive modeling performance of subsequent deep learning models.
- Exhaustive experimental simulations on a case study of Brent crude oil prices demonstrate the efficiency of the proposed solution in modeling over the state-of-the-art solutions from technical, statistical, and computational standpoints. The findings obtained from these simulations are exploited to drive useful implications and insights for sustainable management in crude oil markets.

1.3. Paper organization

The remaining part of this research study is structured as following stated: Section 2 present an analytical overview of the literature studies related to COP prediction. Section 3 argues the working methodology of the proposed framework. The environmental setting of the conducted experiments is described in Section 4. More, the discussion of results and analysis is described in Section 5. The implication of this work is presented in Section 6. Section 7 concludes this study and presents the

possible future direction.

2. Related work

This section discusses the literature studies related to the predictive modeling of prices in crude oil markets and financial trading. The discussion of related works follows the before mentioned categorization of COP prediction methods.

2.1. Econometric methods

The econometric methods generally emphasize using a combination of statistical tools and economic principles to approximate the economic variables and to estimate the target variables. For example, (Moshiri & Foroutan, 2006) consisted of GARCH and ARIMA to forecast COP. In (Gao, Aamir, Shabri, Dewan, & Aslam, 2019), the COP data is decomposed into multiple modes leveraging ensemble empirical mode decomposition (EEMD), where average mutual information (AMI) was followingly used to reconstruct the data into stochastic and deterministic elements. Then, the ARIMA model was used to model price patterns in the stochastic and deterministic data element, while Kalman Filter (KF) was used to define the input parameters. Additionally, (Y. Chen, Zhou, Wu, & Li, 2019) applied the complementary EEMD to reduce the end impacts and mode mixing by decomposing the original COP time series into intrinsic mode functions (IMFs). Then, they applied ARIMA and sparse Bayesian learning with shared kernels to forecast the target values for the rest and each particular IMF individually. The final predictions are computed by adaptively selected according to the training precision. Again, the ARIMA model was applied in (Chai et al., 2019) to forecast price from reconstructed components from decomposed COP time-series. Moreover, (Marchese, Kyriakou, Tamvakis, & Di Iorio, 2020) studied the correlation between the COP and its developed products by investigating fractionally integrated multivariate GARCH methods for COP forecasting in different oil-related markets and analyzing their performance from a risk management standpoint. FURTHERMORE, (Xie, Liu, Qian, & Li, 2021) investigated the relationships between the Chinese stock market and global crude oil market by applying a BEKKGARCH method to the data decomposed with an adaptive-Fourier-decomposition method. The findings demonstrate that the spillover impact between markets uncovered by BEKK-GARCH can help disclose the relationships between markets under outer shocks with detailed information regarding market oscillations.

Remarkably, the operational characteristics of conventional methods (e.g., ARIMA, GARCH, BEKKGARCH, ARFIMA, etc.) are somewhat static and easy; the stricter hypotheses are applied to the financial data such as COP time-series in our case. Though the time series data meeting the beliefs are well-tailored to analytical performance, these hypotheses are frequently different from the attributes of real-world COP data, which is known to be nonlinear, non-stationary, extremely complicated, contain volatile, and contain structural breaks.

2.2. Machine learning methods

In a stream of solutions to address the shortcomings of traditional Econometric methods, the research attention moves toward ML approaches owing to its ability to automatically learn embedded patterns in data. With the availability of financial time series, many research efforts have been devoted to investigating a variety of ML algorithms to deliver intelligent decisions from such data. In this spectrum, (Jue Wang, Zhou, Hong, Li, & Wang, 2020) presented a multi-granularity heterogeneous integration ML approach to enhance the prediction performance from COP, where seven feature selection methods are employed to recognize key features that influence COP and build numerous granular areas. Then, the obtained distinctive feature spaces are included to generate predict future COP using three common ML predictive models namely SVR, ANN, and Linear regression (LR). The

final predictions are computed by merging the forecasts from the above models in different granular spaces, where an artificial bee colony (ABC) is leveraged to calculate the optimal weighting vector. In addition, (Butler, Kokoszka, Miao, & Shang, 2021) studied the COP predictions under economic and market conditions by introducing a focused time-delay neural network to accurately predict the term structure more over all periods of a different number of months ahead. They replaced the Nelson-Siegel curves with B-spline to result in 1 and 3 months ahead of predictions contrasted to convergence-centered predictions. Moreover, (Deng et al., 2019) studied the Trading in Crude Oil Markets by integrating d Non-Dominated Sorting Genetic Algorithm II (NSGA-II) and Multi-Class Support Vector Machine (MCSVM) in a single approach for predicting and simulating the trading in two famous crude oil markets. The MCSVM was applied to predict the development direction and significance stages and NSGA-II was used to adjust the threshold rates of trading regulations. More, (Yu, Xu, & Tang, 2017) used the grid method to generate user-defined parameters for Least-squares SVR (LSSVR) based on probability distributions. Then used random sets of generated parameters to designate different individual LSSVR participants for predicting COP. The final forecasts were obtained ensemble weighted averaging of different participants. Furthermore, kernel extreme learning machine (KELM) was employed in (Zhang et al., 2021a) for price forecasting from COP data decomposed using two subsequent decomposition methods, variational mode decomposition (VMD) decomposing the original COP data and EEMD for decomposing the residual term containing multifaceted information. Despite the success achieved by ML algorithms, their performance hangs on the quality of feature engineering applied and their performance worsens with increased complexity and irregularity of COP data.

2.3. Deep learning methods

Deep learning, as an outstanding choice, is a promising direction for COP prediction because of its brilliant capability to model complicated nonlinear time series. Hence, it turned to gain the research attention for the deterministic and probabilistic predictive modeling different forms of financial time-series, which empower and automate the decision-making and risk management with little human interventions. In this regard, for example, (Niu, Wang, Lu, Yang, & Du, 2020) introduced a deep learning model built by stacking convolutional layers and GRUs framework to model temporal patterns from the decomposed COP time-series using VMD. Then, their model was evaluated for deterministic and probabilistic predictive modeling. In addition, (Busari & Lim, 2021) presented also studied and compared the performance of GRU and LSTM for COP prediction. Besides, (Y.-C. Chen & Huang, 2021) investigated the role of CNN and LSTM in exploiting financial indicators and characteristics (i.e., COP, gold prices, volatility indices) to precisely forecast future stock price trends. Moreover, (Urolagin, Sharma, & Datta, 2021) addressed nonlinearity and complexity problems in COP time-series by using six deviations of multivariate LSTM (MLSTM) models with various features and input transformations including Z-score transformations. Furthermore, (Karasu & Altan, 2022) used LSTM to model COP data based on volatility indicators (including Kaufman's adaptive moving average (KAMA), simple moving average (SMA), an exponential moving average (EMA)), trend indicators (including highest high-lowest low (HHLL), volatility ratio (VR), average true range (ATR)), and momentum indicators (including the rate of change (ROC), commodity channel index (CCI), and relative strength index (RSI)), while chaotic Henry gas solubility optimization is leveraged for hyper-parameter optimization. CNN's have also been investigated for predictive modeling in some energy-related tasks but not for COP prediction (Geng et al., 2020). For example, temporal Convolutional Networks (TCN) are presented in (Lara-Benítez, Carranza-García, Luna-Romera, & Riquelme, 2020) as a specialized network with improvements over recurrent networks in performing predictive modeling from energy-related data. It enables the extraction of long-term representations utilizing residually

connected dilated causal convolutions (Yin & Xie, 2021). A common limitation of these studies is they did not consider the statistical and dynamic characteristics of data during the training. Despite the success of DL in many fields; however, its applicability in Crude oil markets and trading is still in infancy, which necessitates more investigations in this direction.

2.4. Hybrid methods

The hybrid methods refer to the studies that attempt to build a predictive model by combining one or more techniques from the above-mentioned categories aiming to take the advantage of different constituting techniques. In this context, (Bildirici, Guler Bayazit, & Ucan, 2020) presented a hybrid framework that integrates LSTM and Logistic Smooth Transition Autoregressive Generalized Autoregressive Conditional Heteroskedasticity (LSTARGARCH) for analyzing volatility COP under the impact of the COVID-19 pandemic. Besides, (Abdollahi & Ebrahimi, 2020) presented a hybrid approach for capturing linear and non-linear characteristics of the COP time series. This hybridization was performed by combining ARFIMA, Adaptive Neuro-Fuzzy Inference System (ANFIS), and Markov-switching models with identical weights, generated with multi-objective genetic algorithm to guarantee the robust hybrid model with the minimum prediction error. In addition, (Jianzhou Wang, Niu, Du, & Yang, 2020) proposed a hybrid ensemble probabilistic prediction framework that integrates a back-propagation neural network (BPNN), LSTM, Gaussian process regression (GPR), least absolute shrinkage and selection operator (Lasso), and bidirectional LSTM (Bi-LSTM), where the probabilistic forecasting periods generated by these models were aggregated according to the model weights computed an adaptive grey wolf optimizer with Cuckoo search (AGWOCS). Moreover, (J. Liu & Huang, 2021) presented a dual-stream framework, called AGESL, for predicting COP based on historical prices and textual news. The AGESL employed the ARIMA model to predict the mean price of COP, and handle volatility prediction using the GARCH model, where LSTM was used to learn the long and short historical representations included in the news events and prices. Furthermore, (Abdollahi, 2020) presented a hybrid solution for COP forecasting that integrated completed CEEMD for decomposing COP data over the time dimension into two types including nonlinear and volatile series. Then, the nonlinear series are forecasted by the SVM optimized with particle swarm optimization (PSO), while volatile series was predicted by Markov-switching (MS) GARCH (MS-GARCH). The study (Busari & Lim, 2021) shows that integration of the AdaBoost algorithm and GRU help improve the COP prediction performance.

3. Materials and analysis

This study considers Europe Brent spot price data, gathered from the United states' Energy Information Administration, as a case study for evaluating and analyzing the performance of the proposed framework. The price data is given in terms of dollars per barrel. The daily price data were collected from May 20, 1987, to January 3, 2022. The weekly data were recorded from May 15, 1987, to December 31, 2021. The weekly data is calculated by averaging the price of days of weeks, where each week is chosen to end with Friday. As a simple preprocessing step, the data was cleaned by dropping the null or missing values to help analyze and use the data in our experiments.

After that, daily and weekly studied data undergo statistical analysis using statistical distribution using the maximum likelihood estimation to discover the best fitting statistical distributions. Among ninety statistical distribution, the top three fitting distribution functions are as follow: Singh-Maddala distribution, alpha distribution, and fatigue life distribution in order of their fitting performance. The distribution plots and the fitting curves of these distributions are displayed in Fig. 1. In similar way, three distribution functions provide best fitting performance on weekly data namely alpha distribution, fatigue life distribution, and in gauss distribution, in order of their fitting performance (i.e., best fitting comes at first). The distribution plots and the fitting curves of these distributions are displayed in Fig. 2. It is worth noting that the indicated subplots (in Fig. 1 and Fig. 2) shows the statistical properties of the case study data with a greater fitting performance in decreasing order (i.e., from left to right). In Table 1, the parameters of the above-mentioned distribution functions for both daily and weekly COP data are presented.

4. Methodology

In this section, we introduce the building and the working methodology of the proposed framework for predictive modeling of COP in Oil markets. In particular, the proposed approach consists of four main constituent components namely Optimized Variational Mode Decomposition (OVMD) for hyperparameters optimization, AdaBoost Random Forest (ARF) for predictive modeling of high frequency series, and multi-path spatial-temporal DL model for predictive model of low-frequency series. The following subsections dive into the details of each of these components. The proposed predictive modeling framework is shown in Fig. 3.

4.1. Optimized variational mode decomposition (OVMD)

Decomposing a complicated non-linear COP time-series into multiple

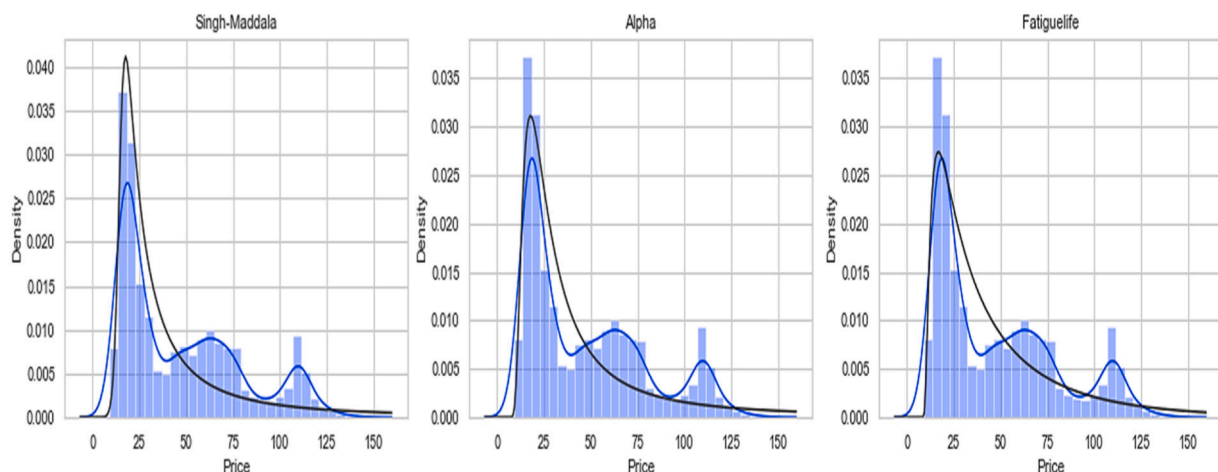


Fig. 1. Statistical distribution fitting for the daily Europe Brent spot price data.

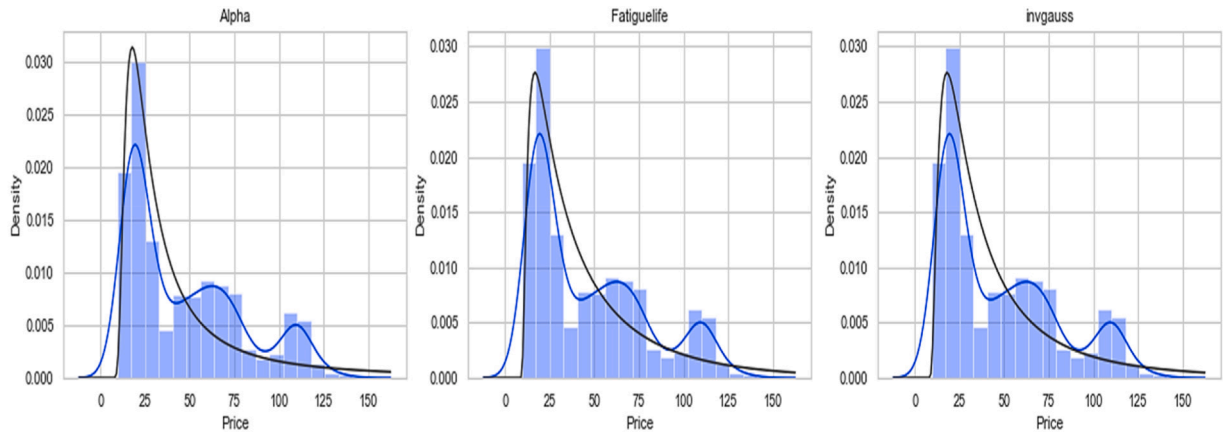


Fig. 2. Statistical distribution fitting for the weekly Europe Brent spot price data.

Table 1

The parameters of the nine statistical distribution functions with best fitting performance on Brent COP data.

Distribution	Parameters	Values	Distribution	Parameters	Values
Daily COP data			Weekly COP data		
Singh-Maddala	Shape	9.43849	Alpha	Shape	0.29387
	Location	0.11714		Location	3.37643
	Scale	15.3614		Scale	22.6078
Alpha	Shape	0.37073	Fatiguelife	Shape	1.01422
	Location	3.04751		Location	7.86011
	Scale	23.9058		Scale	25.8162
Fatiguelife	Shape	1.00621	Invgauss	Shape	1.11492
	Location	7.74767		Location	6.85402
	Scale	25.9749		Scale	36.1253

intrinsic mode functions, with the intention of eliminating the unnecessary elements embedded in the time-series, is of great importance for predictive performance and the generalization capabilities of DL models. EMD-based decomposition generally suffers from mode overlap, sensitivity to noise, boundary effects and absence of precise mathematical details. These restrictions might impact the predictive modeling capabilities of the DL model results. As a remedy, Variational mode decomposition (VMD) (Konstantin & Dominique, 2014) was proposed as an iterative, multiresolution, (non-recursive) technique for time-series decomposition, that shows broad acceptance in the domain of signal processing, fault diagnosis (Zhang et al., 2021b), energy systems (Zhang et al., 2021a), and electrocardiogram evaluation (Goovaerts et al., 2018; Y. Liu, Yang, Huang, & Gui, 2020). VMD methods are mainly depending on Hilbert transform and Wiener filtering for automatic searching for the ideal solution of the problem of constrained variations by decreasing the complete bandwidth of all modalities. Among other decomposition techniques, the VMD has many advantages (Konstantin & Dominique, 2014) over other time-series decomposition techniques. First, it can effectively deal with noisy data with high uncertainty. Second, it can efficiently eschew the recursive sifting dilemma. Third, it can adaptively specify the relevant filter limits.

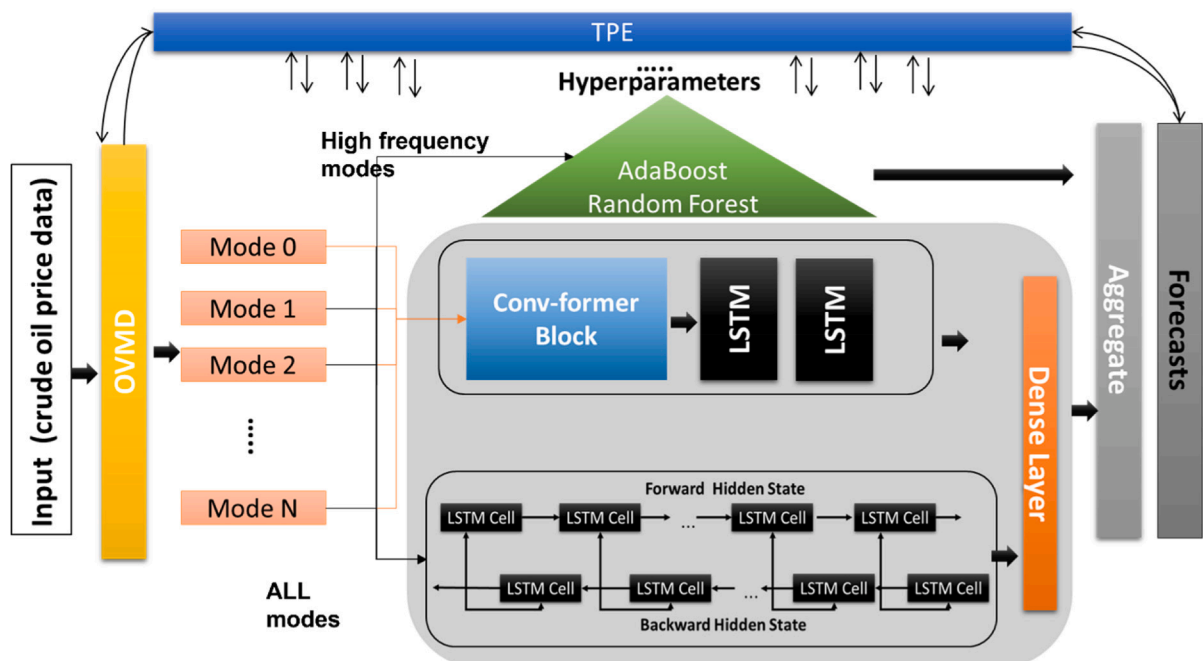


Fig. 3. Illustration of the internal structure of the proposed framework for predictive modeling of COP data.

By applying VMD to COP data, the original time-series $f(t)$ could be decomposed into a set of separate modes (or subsequence) u^i that has a narrow bandwidth. The primary phases of VMD computation are as follows:

- For each subsequence u^i , the Hilbert transform is computed to acquire its specific one-sided field.

$$\left(\delta(t) + \frac{j}{\pi t}\right) \otimes u^i(t), \quad (1)$$

- An exponential tuning indicator $e^{-jw^i t}$ is leveraged to be multiplied by each subsequence u^i to approximate the relevant base frequency w^i , and the mode spectrum is adjusted to the respective baseband.

$$\left[\left(\delta(t) + \frac{j}{\pi t}\right) \otimes u^i(t)\right] \bullet e^{-jw^i t}, \quad (2)$$

- For each subsequence u^k , the projected bandwidth is attained by applying Gaussian softness demodulation. To be specific, the computation task of the automated non-recursive variational decomposition is defined as follows:

$$\min_{\{u^i\}, \{w^i\}} \left\{ \sum_i \left\| \partial^t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \otimes u^i(t) \right] \bullet e^{-jw^i t} \right\|_2^2 \right\}, \quad (3)$$

Subject to,

$$\sum_i u^i = f(t), \quad (4)$$

where $\{u^i\} = \{u^1, \dots, u^i\}$ denote decomposed modal elements, and $\{w^i\} = \{w^1, \dots, w^i\}$ represent the respective base frequencies, the number of decomposed subsequences is i , u^i denotes the i -th subsequence in $f(t)$, $\delta(t)$ denotes the Dirac distribution, $\otimes$ is a convolution operator, and ∂^t represent the partial derivative.

- To attain an ideal solution of the constrained variational mode, the Lagrange multiplier λ and quadratic penalty α are launched to transform the problem into an unconstrained problem formulated by the Lagrange function as follow:

$$L(\{u^i\}, \{w^i\}, \lambda) \left\{ \alpha \sum_i \left\| \partial^t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \otimes u^i(t) \right] \bullet e^{-jw^i t} \right\|_2^2 + \|f(t) - \sum_i u^i(t)\|_2^2 + \left[\lambda(t), f(t) - \sum_i u^i(t) \right] \right\}, \quad (5)$$

where α denotes the stabilizing parameter of the data-fidelity control. Following, the Lagrange calculation of u^i is transformed from time-domain to frequency-domain using Fourier transform, and the saddle points are computed via the alternate direction method of the multiplier by consecutively updating $\tilde{u}_{n+1}^i$, $\tilde{w}_{n+1}^i$ and λ_{n+1} .

$$\frac{\tilde{u}_{n+1}^i(w)}{1 + 2\alpha(w - w^i)} = \tilde{f}(w) - \sum_{k \neq i} \tilde{u}_n^k(w) + \frac{\tilde{\lambda}(w)}{2} \quad (6)$$

$$w_{n+1}^i = \frac{\int_0^\infty w |\tilde{u}^i(w)|^2 dw}{\int_0^\infty |\tilde{u}^i(w)|^2 dw}, \quad (7)$$

$$\tilde{\lambda}_{n+1}(w) = \tilde{\lambda}_n(w) + \tau \left[\tilde{f}(w) - \sum_i \left| \tilde{u}_{n+1}^i(w) \right| \right], \quad (8)$$

where represent the iteration number, $\tilde{u}_{n+1}^i(w)$ denote the Wiener filter of the existing residual $\tilde{f}(w) - \sum_i |\tilde{u}^i(w)|$; $w_{n+1}^i(w)$ represent the spec-

trum's base of gravity of the existing mode. $\tilde{u}^k(w)$, $\tilde{u}_{n+1}^i(w)$, $\tilde{f}(w)$, and $\tilde{\lambda}(w)$ denote the Fourier transform of the $u^k(w)$, $u_{n+1}^i(w)$, $f(w)$, and $\lambda(w)$, respectively. τ denote the upgrade step.

In the nutshell, the VMD can be considered as a function $VMD()$ that decompose the COP time-series $f(t)$ into a group of modes $\{u^i | i = 1, 2, \dots, i\}$, and could be formulated as:

$$u^i(t) | \{i = 1, 2, \dots, i\} = VMD(f(t), i), \quad (9)$$

where i denote the number of generated modes as a pre-determined parameter. Nevertheless, in functional applications, the input time-series are generally discrete. Thus, given n as the discrete-time script, the previous formula can be written as:

$$u^i[n] | \{i = 1, 2, \dots, i\} = VMD(f[n], i) \quad (10)$$

Deterministic predictive modeling often generates typically generates single-point projections with particular preferences, which could be further enhanced by applying denoising techniques and other approaches. The main parameters of the VMD include a number of modes, tolerance ϵ , moderate bandwidth, Initialization method, and Noise tolerance.

To precisely decompose the COP time-series using the VMD, find exact bandwidth and base frequency for each component, and avoid the problem of under-decomposition or over-decomposition, it is required to optimize the main influential parameters of VMD such as a number of modes k , tolerance, and bandwidth.

Accompanying VMD with automated optimization parameters engenders the term OVMD. A broad range of evolutionary algorithms has been investigated in this regard, such as PSO (MA, TONG, & ZHANG, 2018), genetic algorithms, and so on. However, they suffer from complicated search paths, fall into a local optimum, slow convergence, computationally exhaustive. To this end, TPE optimization is employed to optimize the VMD's parameters to maximize the objective function for improving the predictive modeling performance through an iterative search of parameter space. This can tackle the limitations of complicated computation processes and low precision of parameter optimization algorithms while exploiting global and local search abilities. Therefore, the TPE-based parameter optimization algorithm is applied to search for an ideal set of parameters of the OVMD that can empower the proposed framework to reach the ideal predictive modeling for COP data. A detailed discussion about TPE optimization is provided in a later section.

By applying OVMD to decompose daily and weekly Brent COP time-series, we obtain a set of nonlinear and explosive modes of price information as shown in Fig. 4. It is worth noting that the components $\{Mode0, Mode1, Mode2\}$ comprise volatile estates of Brent oil prices, whilst the components $\{Mode3, Mode4, Mode5, Mode6\}$ incorporate nonlinear attributes of prices. In this part, the OVMD is applied to decompose the daily and weekly COPs into multiple modes before being fed into training procedures.

4.2. AdaBoost Random Forest

This section discusses the first learner of the proposed framework, in which we introduce AdaBoost Random Forest (ARF) regressor to model the highest frequency modes obtained from the OVMD i.e., $Mode0$ and $Mode1$.

Adaptive Boosting method, which is abbreviated as AdaBoost broadly adopted ensemble learning approach in which weak learners are boosted by cultivating their parameters to perform voting to create a joint model (Freund, 1997). When many ML algorithms are combined into a single forecasting model, this method can decrease the amount of bias. During the boosting operation, the increased weighting of the observations is used to ensure that some of them appear in the new permutations on a consistent basis. Additionally, each classifier is trained on data by taking into account the achievement of the preceding

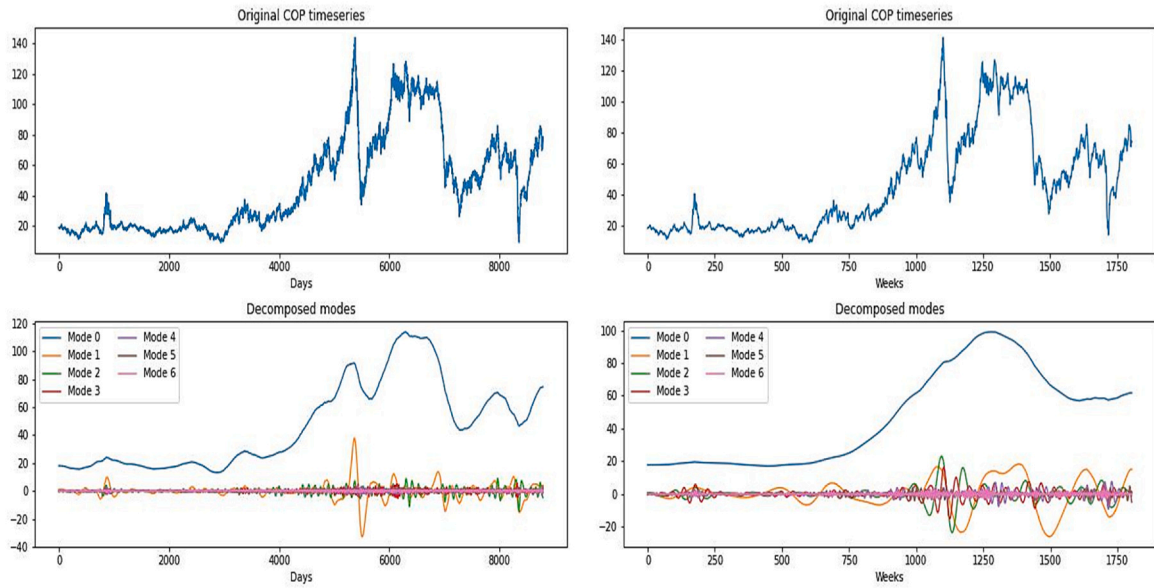


Fig. 4. Visualization of the Brent COP data before and after passed to OVMD. A) daily data at left side; B) weekly data at right side.

classifiers. The weights are adjusted after each training iteration. To draw attention to the most problematic instances, misclassified data is given a higher weighting, then successive learners will be more focused on them.

Our first step is to enter a sequence of m samples, such as $(x_1, y_1) \dots (x_m, y_m)$, where the output “ y ” is such that $(y \in R)$. Random Forest (RF) regressor has been demonstrated as a powerful predictive learner because of its ability to syndicate multiple decision trees in forecasting the output instead depending on one decision trees resulting into non-biased prediction with low variance. Motivated by that, the RF regressor is also used to build a base weak learner for AdaBoost. The training sample set’s sampling weights (W_i) are be initialized. The created model will be used to put in a scikit-learn wrapper and finally boosted. The following are the steps: Initialize the training sample’s sampling weights W_i . Weighing in at the same amount on either side, we find that:

$$W_i = \frac{1}{N}, i = 1, 2, 3, \dots, N \quad (11)$$

Training of Random Forest by training samples and computation of error following the second phase of AdaBoost Algorithm (Blais, 2007) requires N observations for both Random Forest. For $m = 1$ to M . By minimizing the weighted error function F_m , create a classifier $H_m(x)$ based on the training data. Where F_m is computed as follow:

$$F_m = \sum_{i=1}^N W_i^m I[y_i \neq H_m(x_i)] \quad (12)$$

$$\text{Compute err}_m = \frac{F_m}{\sum_{i=1}^N W_i^m} \quad (13)$$

$$\text{Evaluate } \alpha_m = \log\left(\frac{1 - \text{err}_m}{\text{err}_m}\right) \quad (14)$$

Then change the sampling weights by setting.

$$W_i^{(m+1)} = W_i^m \exp[\alpha_m * I[y_i \neq H_m(x_i)]] \quad (15)$$

Steps 2 and 3 should be repeated until all of the network’s predictors have been acquired.

As the final step of the AdaBoost Algorithm, output the predicted result by combining the predicted results of all predictors with ensemble weights.

$$H_m(x) = \operatorname{argmax} \left[\sum_{m=1}^M \alpha_m H_m(x) \right] \quad (16)$$

RF regressor is a common ML (M. Learning, 2017) that constructs numerous unconnected decision trees using the random resampling bootstrap methodology and node stochastic regression algorithm and then votes on the final regression outcomes. Features with complicated relationships can be analyzed using RF. The RF technique is more resistant to noise and incomplete information, as well as has a quicker learning velocity. For massive and large-scale data, the parameter - importance assessment can be utilized as a feature choice method. (Joelsson, Benediktsson, & Sveinsson, 2006; Strobl, Boulesteix, Kneib, Augustin, & Zeileis, 2008) Both goals of RF feature choice are to locate extremely reliant feature parameters and to locate feature parameters - with minimal data and a good capacity to represent predicting outcomes. There are three phases to compute the importance of every feature y_m , where f features of $y_1, y_2, y_3, y_4, \dots, y_f$. First, the bootstrap approach is used to randomly take N modern self-help instance sets from the main training dataset, then reuse them to build K trees of regression. Second, a features ($a \leq b$) are randomly obtained as a randomly produced feature subset at every node of every tree, and by determining the data embedded in every feature that is embedded in this feature subset, the better regression capacity between the a features are choice for node dividing, raising the uniqueness of decision trees. Third, we can compute the feature relevance in the tree as:

$$\text{IMP}_{km} = \sum_{a \in A} \text{IMP}_{ma} \quad (17)$$

Where k refers to the decision tree. The y_m is a feature locate in k fits to set k . We can compute the relevance feature in B trees as:

$$\text{IMP}_m = \sum_{k=1}^B \text{IMP}_{km} \quad (18)$$

We can compute the normalized scores of relevance as:

$$\text{IMP}_m = \frac{\text{IMP}_m}{\sum_{m=1}^f \text{IMP}_m} \quad (19)$$

4.3. Spatial-temporal representation learning

This section discusses the second learner of the proposed framework, in which we introduce multi-path DL regressor to model the inherent representations from both low and high frequency modes obtained from the OVMD. The goal of predictive modeling of COP time-series is to provide precise and fast price information for the near future in order to improve price management proficiency in oil markets. Because crude market conditions are typically unpredictable and nonlinear, a good prediction model should be able to incorporate and handle such complexities. To increase prediction performance, many statistical or ML-based predictive models have been developed. Representational learning is a critical step in developing an effective predictive model that extracts and selects the most representative characteristics from past COP data. COP data features typically demonstrate a spatial-temporal relationship and a periodic characteristic. More specifically, pricing information is influenced not just by market conditions but also by preceding times. Furthermore, the COP time-series reveals a daily or weekly periodic rhythm. All of these characteristics of COP data, when combined, determine the predicted future price of crude oil in either local or global markets. To this end, we build a novel DL model to exploit both spatial and temporal relationships representations for the decomposed COP data.

Spatial Representation Learning (SL) module: It has been proven that convolutional layers and networks can extract spatial features from a variety of data sets. At extremely low computational constraints, even the most powerful and efficient CNNs begin to degrade significantly in performance. This can be attributed to the fact that the width and depth of the network are constrained due to very low computation resources, which is critical for network performance but proportionate to the expense of calculation. To achieve a better trade-off between the performance and computational cost, common approaches have been applied to make the networks “wider” or “deeper”. Another limitation of the existing CNN solutions is that they fail to capture local and global patterns in the input sequences, which limits their predictive modeling capabilities. Nevertheless, they both lead to high computation costs, thus are not welcoming to the efficient DL model. To address this issue. We propose a Conv-former block that expands neither the depth nor the width of the network but empowers the representation power of the network by accumulating a group of K concurrent convolutional kernels as an alternative to traditional in which one kernel is applied at layer (see Fig. 5).

The calculations in the convolutional kernel can be formulated as follows:

$$Y_t^i = A(W^i * X_t^i + b^i), \quad (20)$$

whereas W^i and b^i denote the weight and bias parameters, X_t^i represent the input sample at time t . The convolution operation is denoted as $*$, while the activation function is denoted as A . The kernel is dynamically designated to aggregate K linear functions:

$$\begin{aligned} \tilde{W}(x) &= \sum_{K=1}^K \pi_K(x) \tilde{W}_K, \quad \tilde{b}(x) = \sum_{K=1}^K \pi_K(x) \tilde{b}_K, \\ \text{s.t. } 0 &\leq \pi_K(x) \leq 1, \quad \sum_{K=1}^K \pi_K(x) = 1, \end{aligned} \quad (21)$$

while the linear function $W_K^T x + \tilde{b}_K$ has a weight of transformer attention of type K , denoted by the symbol π_K . If you notice, the weight $\tilde{W}(x)$ and the bias $\tilde{b}(x)$ are both functions of input and share the same attention. For each input x , the transformer attention $\{\pi_K(x)\}$ is computed in a non-constant way. For a given input, they reflect the best combination of linear models $\{\tilde{W}_K^T x + \tilde{b}_K\}$. This way, the convolutional kernel is more robust in terms of representational power than the standard convolu-

tion.

The transformer attention block $\{\pi_K(x)\}$ is implemented using a couple of fully connected layers (FCLs), convolutional 1D layer, as well as the Multi-Headed Self-Attention (MHSA) layer (Vaswani et al., 2017), as illustrated in Fig. 5. Unlike transformer network (Vaswani et al., 2017), our block use two half-steps FCLs located prior to and after the MHSA layer, where half-step residual connection is used to improve gradient flow and avoid vanishing problems. By the end of the second FCL, a layer-norm layer is introduced to re-normalize the learned maps to accelerate computations. The output y_i can be delivered from the input x_i of transformer attention block as follows:

$$\tilde{x}_i = x_i + \frac{1}{2} FCL(x_i) \quad (22)$$

$$x'_i = \tilde{x}_i + MHSA(\tilde{x}_i) \quad (23)$$

$$x''_i = x'_i + CONV - 1D(x'_i) \quad (24)$$

$$y_i = LayerNorm(x''_i + \frac{1}{2} FCL(x''_i)) \quad (25)$$

Dropout layer is applied to avoid the overfitting issues. There's no pooling operation applied following Conv-former in our model because the dimension of spatial representation maps isn't very large. Following, the output of Conv-former is subsequently fed into an LSTM network.

Temporal Representation Learning (TL) module: COP time series are known to display temporal correlations between neighboring periods. Analysis of the underlying temporal characteristic in sequenced data is often done using RNNs (Géron, 2019). Because it is subject to the vanishing gradient problem, RNN fails miserably when dealing with extended sequences. Long-period interaction in sequential data can be learned using LSTM. An LSTM construction contains three elementary gates namely an input gate, an output gate, and a forget gate, while it uses a memory cell for storing temporal state. The input and forget gates both have an effect on the current state. As an example, LSTM employs a forget gate to decide the amount of information from past cell states to be maintained in the current cell state, and an input gate to decide how much information from input cells needs to be stored in the current cell state (Hochreiter & Schmidhuber, 1997). Gating mechanisms are used in LSTM to control the flow of data. At every time step, the volume of data received that will be kept is calculated methodically. As a result, LSTM may remember temporal patterns more than an extended length of time. A vanilla LSTM memory block's unit and triple gate architectures are shown in Fig. 6.

Increasing the depth of the deep network is a common way of improving its performance motivated by that, multiple layers of LSTM are stacked after Conv-former block to allow us to capture more complex aspects of COP data that empower the temporal learning capabilities of our framework. This is accomplished by passing the hidden state of each layer as an input to the next layer in the stack.

The output of forgetting gate g is computed with respect to time t as:

$$g_t = \sigma(I_{ad_g} a_t + I_{dd_g-1} d_{t-1} + q_{d_g}) \quad (26)$$

The output of input gate n is computed with respect to time t as:

$$n_t = \sigma(I_{ad_n} a_t + I_{dd_n-1} d_{t-1} + q_{d_n}) \quad (27)$$

The outcome of the output gate u is computed with respect to time t as:

$$u_t = \sigma(I_{ad_u} a_t + I_{dd_u-1} d_{t-1} + q_{d_u}) \quad (28)$$

The candidate state c is computed with respect to time t as:

$$c_t = \tanh(I_{ad_c} a_t + I_{dd_c-1} d_{t-1} + q_{d_c}) \quad (29)$$

The cell state s is computed with respect to time m as:

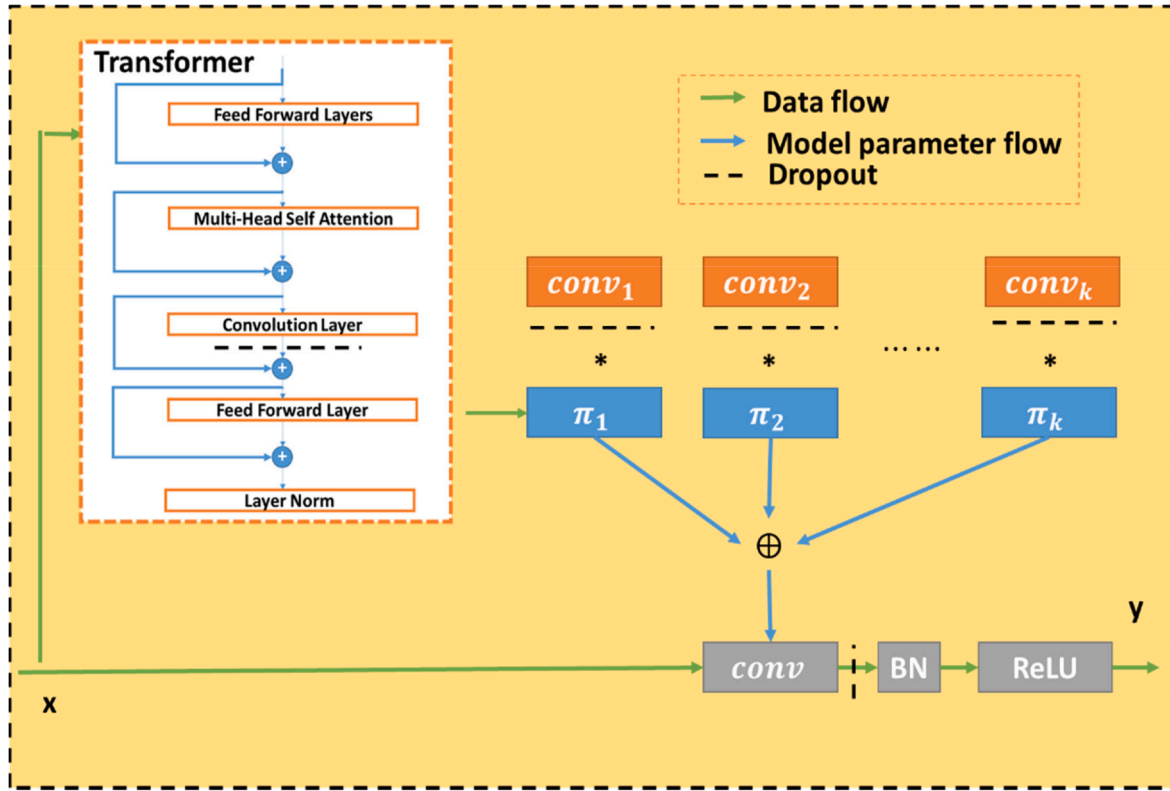


Fig. 5. Illustration of the internal design of the proposed Conv-former module for spatial representational learning.

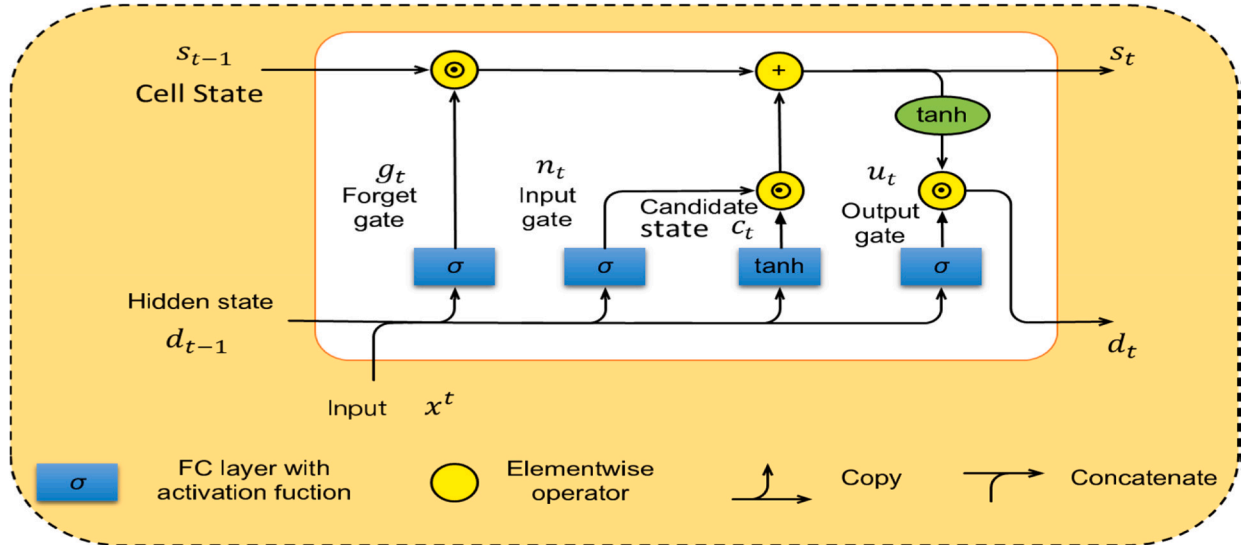


Fig. 6. Illustration of the internal design of the vanilla LSTM for temporal representational learning.

$$s_m = g_t \odot s_{t-1} + n_t \odot c_t \quad (30)$$

The hidden state d_t with respect to time m as:

$$d_t = u_t \odot \tanh(s_t) \quad (31)$$

The element-wise multiplication, hyperbolic tangent function, and sigmoid function are represented by $\odot$, $\tanh(\bullet)$, and $\sigma(\bullet)$. The input is

represented by a_t at time t . The weights and bias are represented by I, q .

First, the weighted total of a_t, d_{t-1} , and the bias is used to determine g_t . The sigmoid function gives it a value in the range of $[0-1]$, with 1 indicating that every information in the input transferred and 0 indicating the reverse. When changing the cell state at time t , the forget gate regulates the quantity of information in the previous cell state at time $t-1$, i.e., s_{t-1} . n_t and u_t are calculated similarly to g_t by implementing

the sigmoid function to the weighted total of a_t, d_{t-1} , and every bias. It also calculates how much incoming information is kept in the cell state and how much of the cell state value will be transferred at time t . The incoming information to be kept in c_t is calculated by applying $\tanh$ to the weighted total of a_t, d_{t-1} , and bias of s_t is changed using the total of s_{t-1} and c_t , which are regulated by g_t . Finally, the component total of u_t and $\tanh$ of s_t is used to get the output value d_t .

Bi-directional Learning (BL) module: It's well-known that the behavior of oil markets tends to follow an irregular pattern over time, which may be reflected in the COP time series. Aside from that, the price information of the preceding time also influences the COP for the following time, which in turn influences the flow of the preceding period. Bi-directional LSTM networks are used to extract periodic features and capture temporal dependency from the historical COP data in order to investigate these qualities. A couple of one-directional LSTMs are stacked up and down in a bi-directional architecture (see Fig. 7), one for the forward direction and the other for the backward direction, respectively. This way, the forward LSTM performs temporal representation learning based on previous price information, while backward LSTM performs temporal representation learning according to the forthcoming price information. In the previously mentioned temporal representation learning, all that's taken into account here is the previous COP information. The hidden states of forwarding LSTM and backward LSTM are eventually integrated to produce the final representational maps. Bidirectional learning help extract more and possibly richer representations from COP data in both directions that inevitably empower the predictive power of our framework.

4.4. Tree-structured Parzen optimization (TPE)

In the creation of ML solutions, automated hyperparameter tuning is critical, particularly when deeper models, such as the proposed framework, are employed. Using it, it is possible to reduce the amount of human labor required to create the model while simultaneously boosting network performance by picking hyperparameter values that are best for the application at hand. In this regard, Sequential Model-based Bayesian Optimization techniques have been showing wide acceptance for hyper-parameter optimization. In comparison to more traditional optimization methods like grid search or RS, the use of BO techniques

has the advantage of requiring fewer function evaluations. The reason for this is that these systems learn and pick the optimum hyperparameter sets based on their distributions representing the fitness scores in the prior rounds of the process. Statistically guided sampling reduces the amount of hyper-parameter samples collected from the search space so that the most interesting options could be evaluated.

Different from standard Bayesian optimization, Tree-structured Parzen Estimator (TPE) has been proposed to deal with conditional and qualitative parameters, and therefore to empower the efficiency of selecting ideal hyperparameters. Hence, it is selected in this work to search for the optimum hyperparameters of the proposed framework at model and decomposition (i.e., VMD) scale. Since it involves fewer function evaluations than other optimization methods like random and grid search, the TPE method is more quickly able to arrive at its optimal performance. Unlike Bayesian Optimization, it has the ability to optimize categorical and conditional hyperparameters affording broader scope hyperparameter options. For constructing probability density functions in the hyperparameter search space, TPE relies on the Parzen-window density estimation (sometimes referred to as kernel density estimation). In particular, the mean and standard deviation of each instance, defines a Gaussian distribution in the search space of the model's hyperparameter, so that the hyperparameter space has a Gaussian distribution.

An arbitrary search is carried out during the earlier iterations, N^{init} , to define the initial distributions by selecting samples from the response surface $\{\theta^i, y^i\} (i = 1, 2, \dots, N^{init})$, whereas θ designates the model's hyperparameter and y denote the equivalent fitness score of the response surface. Follow, the hyperparameter space is split into two parties, specifically *great* and *awful* instances based on a predefined threshold y^* of the fitness value, where the probability distributions Pr^g and Pr^a are used to define these two parties for the set of hyperparameters θ :

$$p(\theta|y) = \begin{cases} Pr^g(\theta) & \text{if } y < y^* \\ Pr^a(\theta) & \text{if } y \geq y^* \end{cases} \quad (32)$$

Next, the TPE calculate the expected improvement (EI) for each step:

$$EI(\theta) = \frac{Pr^g(\theta)}{Pr^a(\theta)}, \quad (33)$$

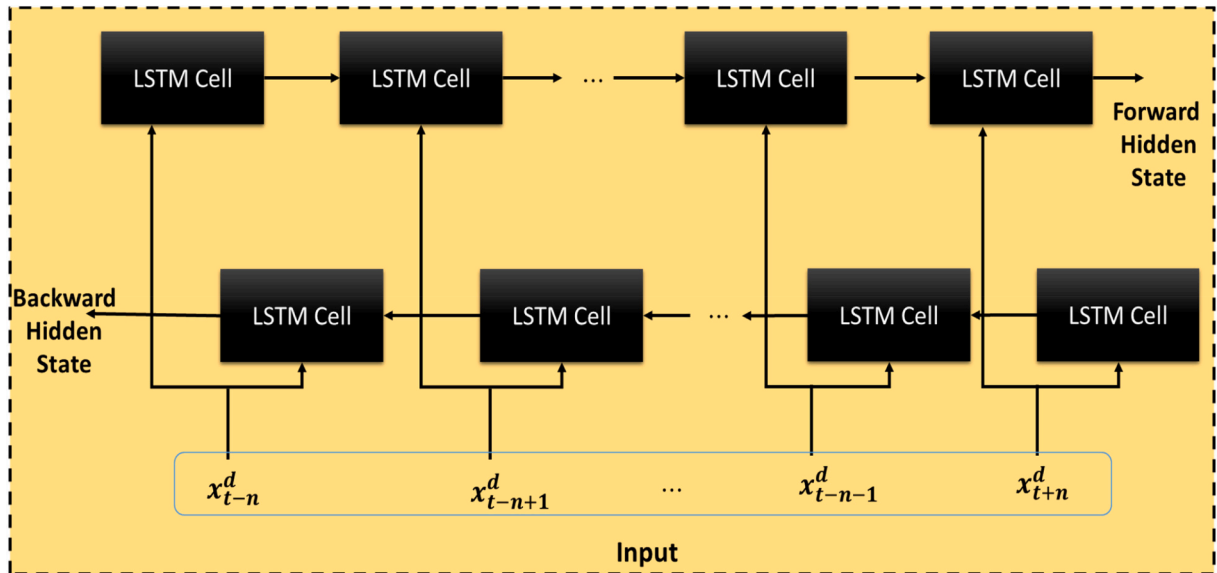


Fig. 7. An illustration of bidirectional learning module built with LSTM.

After that, given maximum El , the TPE algorithm choose hyperparameter setting θ^* . in other words, the TPE carefully chooses the optimum hyperparameters according to the collection of most excellent samples as well as the corresponding distributions, instead of relying only on the best one. The pseudocode of TPE used in our framework is presented in Algorithm 1.

Algorithm 1

: TPE FOR OPTIMIZING HYPERPARAMETERS OF THE PROPOSED FRAMEWORK

Input: Set of hyper-parameters and corresponding search space i.e., # decomposition modes, training optimizers, activations, loss function, learning rate, # AdaBoost estimators, Random Forest parameters

Output: A set of optimal hyperparameters θ^* .

```

1:  $N^{init} \leftarrow$  number of initialization iterations
2: fitness function  $\leftarrow$  RMSE
3: For  $i = 1, \dots, N^{init}$  do
4:    $\{\theta^i, y^i\} \leftarrow$  initialize distribution
5: end for
6:  $t \leftarrow N^{init} + 1$ 
7: Loop until no improvements or max iteration reached
8:   Split hyper-parameters into two great and awful
9:   Calculate  $Pr^g(\theta)$  and  $Pr^a(\theta)$ 
10:   $p(\theta|y) \leftarrow \begin{cases} Pr^g(\theta) & \text{if } y < y^* \\ Pr^a(\theta) & \text{if } y \geq y^* \end{cases}$ 
15:  Maximize  $El(\theta)$ 16:  $t + 1$ 
17: End Loop
18: Return  $\theta^*$ .
```

A multiplicity of hyperparameters of the proposed framework are optimized using TPE algorithm, which we summarize the corresponding distribution and search domain in Table 2. 5-fold cross-validation strategy is followed during the optimization to avoid the overfitting issues in the goal function, where 30 is set as the number of trials of TPE.

5. Experimental environment

This section gives a detailed explanation of environment design aspects related to simulation experiments performed and discussed in this work. In detail, the design of our simulation is deeply figured out in terms of implementation setup of the experiments, evaluation metrics used to assess the predictive modeling performance, data preparation steps applied before training.

5.1. Implementation setup

The implementation of the proposed solution and competing methods are coded in Python 3.8.1 making use of the Keras 2.4.3 API

Table 2

The hyperparameters optimized by tpe in the proposed framework.

Hyperparameter	Type of distribution	Domain
Number of Modes (OVMD)	Uniform	[2, 18]
Bandwidth (OVMD)	Uniform	[1000, 5000]
Tolerance (OVMD)	Uniform (float)	[1e-3, 1e-7]
Number of estimators (ARF)	Categorical	[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
Optimizer	Categorical	{SGD, Adam, RMSprop}
Activation	Categorical	{Tanh, ELU, ReLU, LReLU}
Learning rate	Uniform (float)	[0.0001, 0.1]
# Attention heads	Uniform	[1, 12]
Filters	Categorical	[32, 64]
kernel size	Categorical	[3, 5]
Dropout rate	Uniform (float)	[0, 0.7]
LSTM layers	Categorical	[16, 24, 32, 48, 64]

installed on the Tensor Flow 2.3.0 backend. Overall experiments are performed on Windows 10 64-bit OS installed on a Dell workstation armed with an Intel (R) Xeon (R) CPU E5-2670 0@ 2.60 GHz processor, NVIDIA Quadro as a graphical processing unit (GPU), and 256 GB of memory. The statistical analysis is performed using Scipy¹ (Virtanen et al., 2020).

5.2. Evaluation metrics

This subsection describes the performance metrics adopted to evaluate the predictive capabilities of the proposed framework. Specifically, Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Square Error (MSE), Root Mean Square Error (RMSE), Coefficient of determination (R^2), Explained variance score (EVS), and Improvement Rate (IR) are used as follow:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \tilde{y}_i| \quad (34)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \tilde{y}_i}{y_i} \right| \times 100\% \quad (35)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \tilde{y}_i)^2 \quad (36)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \tilde{y}_i)^2} \quad (37)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \tilde{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (38)$$

$$EVS = 1 - \frac{var(y - \tilde{y})}{var(y)}, \quad (39)$$

$$IR_C = \frac{C_N - C_B}{C_B} \times 100\%, \quad (40)$$

where $\bar{y} = \frac{1}{N} \sum_{i=1}^N (y_i)$ denote the mean of true values, C represent the predictive evaluation metric (MAPE, RMSE, MAE), and IR_C represent the ratio of improvement achieved by the new model N over the baseline model B .

5.3. Data preparation

It could be noted from previous earlier discussions that the Brent dataset did not fit the normal distribution, contain many empty or null values of price at official holidays, and the price values fluctuated over time dimensions. These kinds of issues are broadly identified to complicate or even worsen the learning process. Hence, The Brent price data is preprocessed by removing the data instance with null price values to alleviate any discontinuity in data. illustration of null removed data is shown in Fig. 4.

Then, the remaining part of data is normalized to have a zero mean μ and unit variance σ as follow:

$$X_{norm} = \frac{x - \mu}{\sigma}, \quad (41)$$

After that, the data is divided into two sets namely the training set containing the first 80 % of data, and the test set containing the later 20 % of data. The exploratory analysis (such as stationary analysis,

¹ <https://scipy.org/>

Table 3

Findings obtained from the augmented dickey-fuller unit root test on daily brent cop data.

Series	Original	Decomposed						
		Mode0	Mode1	Mode2	Mode3	Mode4	Mode5	Mode6
t-statistics	-2.121	-2.623	-11.457	-19.985	-24.067	-21.889	-24.241	-27.731
p-value	0.236	0.088	0.000	0.000	0.000	0.000	0.000	0.000
Critical values								
Confidence Interval 99 %	-3.431097							
Confidence Interval 95 %	-2.861870							
Confidence Interval 90 %	-2.566945							

Autocorrelations analysis, trending analysis) is performed on the whole data, while the prediction results (comparative results, statistical analysis, ablation analysis) are reported on the test set. Again, the results on test are obtained from a fivefold cross-validation is used for evaluating the test results. To avoid the temporal dependence of time series in train and test samples, and stops information leakage, gaps were introduced between train-test splits and in each data fold.²

6. Results and discussions

Stationarity represents a major characteristic that has to be taken into account in the predictive modeling of time series tasks. Since the proposed framework takes as input the volatile modes generated by the VMD to perform predictive modeling, hence, the stationarity check is performed on each mode to validate that the impact of shocks on a stationary sequence would have vanished steadily. In so doing, given the null hypothesis as “data is not stationary”, the Augmented Dicky-Fuller (ADF) test (Dickey & Fuller, 1981) is a common statistical stationarity test, is applied to evaluate stationarity of Brent COP time-series at daily (see Table 3) and weekly (see Table 4) scales. For daily analysis shown in Table 3, the P-value obtained by the ADF test on the original series is 0.236, which is greater than the significance threshold (0.05), implying that the null hypothesis is true, and the COP data is not stationary. Similarly, the same finding is obtained for the first decomposed component {Mode0} generated by VMD, with a P-value of 0.088. The remaining components {Mode1, ..., Mode6} show a P-value less than 0.05, which implies rejecting the null hypothesis and identifying them as stationary components. For weekly analysis shown in Table 4, the P-value obtained by the ADF test on the original series is 0.350, which is greater than 0.05, implying that the null hypothesis is true, and the COP data is not stationary. In a similar way, the same discovery is observed for the first decomposed component {Mode0} obtained from VMD, with P-value of 0.185. The remaining components {Mode1, ..., Mode6} show P-value less than 0.05, which imply rejecting the null hypothesis and identify them as stationary components.

Autocorrelations represent another important decisive factor in predictive modeling of time-series since it signifies the extent to which a time series is interdependent on the historical values. Thus, the Ljung-Box test, whose null hypothesis is “data is independently distributed”, is implemented to find out the autocorrelation in the original series of Brent COP data and the corresponding components generated by VMD at both daily and weekly scales (see Table 5). The number of lags in Ljung-Box test is computed according to the ADF test. On one hand, as shown in Table 5, the P-value obtained by Ljung-Box test on a different type of series is 0.000, which is less than the critical value, implying that the null hypothesis is rejected, and the daily COP data is correlated. A similar conclusion can be obtained for weekly data, where the P-value obtained by Ljung-Box test remain less than the critical value for all types of data series implying intercorrelation between data instances.

For this objective, the Mann Kendall is applied to statically analyze

and determine if the price data contain monotonic increasing or monotonic decreasing tendencies. It is a non-parametric test, whose null hypothesis is “The COP data has no trend”. The result of applying the Mann Kendall test on the daily and weekly Brent COP data are shown in Table 6. For daily data, it is notable that the original price series inherently contain an increasing trend with P-value = 0.000, which is less than the critical value. Similar findings can be obtained for Mode0 of daily data, while other modes can be demonstrated as having no trend. For weekly data, it could be found that the original price series is trendy in nature and similarly Mode0 and Mode1, whose P-values are less than the significance level, which implies rejecting the null hypothesis.

6.1. Comparative analysis

In this section we start evaluating the performance of the proposed framework by performing an end-to-end comparative analysis between the proposed framework and state-of-the-art competing methods namely LR (Jue Wang et al., 2020), SVR (Astudillo, Carrasco, Fernández-Campusano, & Chacón, 2020), DNN (Lago, Marcjasz, De Schutter, & Weron, 2021), RF, LSTM (Y.-C. Chen & Huang, 2021), CEEMDAN RF (Karjadi & Chou, 2022), CEEMDAN LSTM (Zhou, Huang, & Zhang, 2022), CEEMDAN LSTM_CNN (Rezaei et al., 2021), EWT_BILSTM (Jaseena & Kovoov, 2021), VMD_LSTM_ATT_SVR (K. Zhang & Cao), VMD_CONV_GRU (Niu et al., 2020), and CEEMDAN RF_LSTM (Karjadi & Chou, 2022). On one hand, the results obtained from these comparisons on the daily Brent COP data are presented in Table 7. Without loss of generality, the ML solutions (such as LR, SVR, RF) could be observed to achieve the lowest predictive modeling performance possible because of their inability to handle non-linearity and dynamicity of COP data. For example, RF achieves MAPE of 6.042591 and MSE of 19.25443, while SVR obtains MAPE of 10.31477 and MSE of 68.79466. This in turn makes these methods unqualifiable and reliable solutions for empowering the decision-making and risk analysis in oil markets. However, the simplicity of design of these algorithms gives them the merit of being time- and computational-efficient. On the other hand, deep learning methods, such as DNN (Lago et al., 2021) and LSTM (Y.-C. Chen & Huang, 2021), demonstrate acceptable performance in modeling the dynamic characteristics of COP time series. Further, it could be noted that decomposing the COP time-series, using CEEMDAN and VMD algorithms, before passing to DL is beneficial for improving the prediction performance. For example, CEEMDAN LSTM (Zhou et al., 2022) achieve considerable performance improvements over (MAPE:1.8; MSE: 2.6; RMSE: 1.2) over the LSTM (Y.-C. Chen & Huang, 2021). CEEMDAN RF_LSTM (Karjadi & Chou, 2022) show similar observation as the with the previous example. More importantly, the proposed framework shows great predictive modeling performance overcoming all the competing methods in all metrics (MAPE = 0.313241; MAE = 0.141221; MSE = 0.093647; RMSE = 0.306017; R2 = 0.999998; EVS = 0.999876).

For weekly Brent COP data, the results of fair comparative analysis are shown in Table 8. It could be noted that both ML and DL methods show similar behaviors as those observed in the case of daily data. It is worth noting that the obtained result at weekly resolution is generally less than those obtained for daily resolution data. This finding should be

² <http://doi.org/10.5281/zenodo.4707309>

Table 4

Findings obtained from the augmented dickey-fuller unit root test on weekly brent cop data.

Series	Original	Decomposed						
		Mode0	Mode1	Mode2	Mode3	Mode4	Mode5	Mode6
t-statistics	−1.862	−2.260	−4.461	−8.275	−14.949	−16.141	−16.002	−13.151
P-value	0.350	0.185	0.000	0.000	0.000	0.000	0.000	0.000
Critical values								
Confidence Interval 99 %	−3.434031							
Confidence Interval 95 %	−2.863166							
Confidence Interval 90 %	−2.567635							

Table 5

Findings obtained from the ljung-box test on daily brent cop data.

Scale	Series	Original	Decomposed						
			Mode0	Mode1	Mode2	Mode3	Mode4	Mode5	Mode6
DAY	t-statistics	291197.67	324285.34	271144.12	115875.48	97209.87	55374.81	46263.02	39730.49
	Lags	34	37	36	36	37	37	37	37
	P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
WEEK	t-statistics	40303.52	43127.45	33615.27	18527.60	20739.20	17086.47	12793.20	9924.77
	Lags	25	24	21	24	24	24	23	25
	P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 6

Findings obtained from the mann-kendall trend test on daily brent cop data.

Scale	Series	Original	Decomposed						
			Mode0	Mode1	Mode2	Mode3	Mode4	Mode5	Mode6
DAY	t-statistics	79.087	80.638	6.310	1.063	0.527	0.298	0.089	−0.060
	Score	21727491.0	22153337.0	1733577.0	292233.0	144897.0	82105.0	24675.0	−16561.0
	P-value	0.000	0.000	2.786	0.287	0.597	0.765	0.928	0.951
WEEK	t-statistics	35.768	36.929	2.240	−0.330	0.186	0.173	0.138	0.036
	Score	914689.0	943596.0	57248.0	−8448.0	4772.0	4438.0	3542.0	930.0
	P-value	0.000	0.000	0.025	0.740	0.851	0.862	0.889	0.970

Table 7

Comparison of evaluation results of competing methods for daily cop prediction.

Methods	MAPE (%)	MSE	RMSE	MAE	R ²	EVS	Running Time (s)
LR [21]	1.903027	1.746204	1.32144	0.95118	0.990661	0.990661	0.101249
SVR [57]	10.31477	68.79466	8.294255	3.083199	0.632056	0.659692	0.43763
DNN [58]	2.124326	2.173116	1.474149	1.033706	0.988377	0.988379	31.30524
RF	6.042591	19.25443	4.387987	2.000992	0.897019	0.904362	3.46597
ARF	6.039931	19.13007	4.373793	2.006280	0.897284	0.904414	128.7293
LSTM [30]	2.561211	2.844469	1.686555	1.085776	0.984787	0.985435	44.22142
CEEMDAN RF[59]	0.828691	0.336398	0.579998	0.418436	0.998201	0.998205	198.2147
CEEMDAN LSTM [60]	0.694892	0.213141	0.461672	0.351139	0.99886	0.99886	802.5876
CEEMDAN LSTM_CNN [16]	1.064776	0.442687	0.665347	0.527659	0.997632	0.998377	1151.459
EWT_BILSTM [61]	4.001689	7.399007	2.720112	2.117616	0.960427	0.962073	981.4461
VMD_LSTM_ATT_SVR [62]	4.192409	9.371177	3.061238	1.75811	0.949879	0.957917	488.3834
VMD_CONV_GRU [28]	1.006608	0.462632	0.680171	0.507438	0.997526	0.997534	399.7316
CEEMDAN RF_LSTM [59]	0.742087	0.239856	0.489751	0.372101	0.998717	0.998718	741.7804
Proposed	0.313241	0.093647	0.306017	0.141221	0.999998	0.999876	513.9846

taken into account by managers and decision-makers for improving their planning strategies to be built on more accurate forecasts. More importantly, the proposed framework keeps achieving the lowest prediction error overcoming all the competing methods in all metrics (MAPE = 1.123142; MAE = 0.521431; MSE = 0.536142; RMSE = 0.732217; R² = 0.99931; EVS = 0.999859). These findings can demonstrate the competency of the proposed framework among the existing DL solution for predictive modeling of price information in crude oil markets.

To further confirm the efficiency of the proposed framework, the

improvement rates, are computed for each competing method using different performance metrics on both daily and weekly COP data as shown in Table 9 and Table 10, respectively. It is obvious that R² and EVS metrics attain the lowest improvements in either daily or weekly prediction tasks. On the other hand, MAPE, MSE, RMSE, and MAE show great and clear performance improvements. Compared to CEEMDAN LSTM (Zhou et al., 2022), the proposed framework show the lowest IR in the case of daily data (MAPE: 54.922, MSE: 56.063, RMSE: 33.716, MAE: 59.782), and similarly, on weekly data, it achieved the lowest IR (MAPE:14.404, MSE: 23.736, RMSE: 12.671, MAE: 21.538). The above

Table 8

Comparison of evaluation results of competing methods for weekly cop prediction.

Method	MAPE (%)	MSE	RMSE	MAE	R ²	EVS	Running Time (s)
LR [21]	3.599474	5.667119	2.380571	1.784464	0.969665	0.969665	0.039981
SVR [57]	11.26413	67.03153	8.187278	3.822103	0.641195	0.671815	0.102949
DNN [58]	4.300643	8.961614	2.993595	2.010399	0.95203	0.952052	6.003641
RF	7.194767	22.8769	4.78298	2.681774	0.877545	0.886049	0.644006
ARF	7.30649	23.55383	4.853229	2.74198	0.87392	0.8822928	27.7006
LSTM [30]	5.044478	10.25809	3.202825	2.220339	0.945091	0.946017	14.78286
CEEMDAN RF[59]	1.982303	1.69518	1.301991	0.954623	0.990926	0.990977	36.9219
CEEMDAN LSTM [60]	1.312139	0.703007	0.838455	0.664564	0.996237	0.996244	184.7655
CEEMDAN LSTM_CNN [16]	1.528915	0.963278	0.981467	0.798992	0.994844	0.995296	232.7429
EWT_BILSTM [61]	4.946076	10.88492	3.29923	2.559292	0.941735	0.942925	664.4019
VMD_LSTM_ATT_SVR [62]	4.086188	9.403688	3.066543	2.061812	0.949664	0.952131	137.5724
VMD_CONV_GRU [28]	3.445914	5.142929	2.267803	1.676372	0.972471	0.972478	117.6566
CEEMDAN RF_LSTM [59]	1.403212	0.859401	0.927039	0.712317	0.9954	0.9954	165.4378
Proposed	1.123142	0.536142	0.732217	0.521431	0.99931	0.999859	243.1231

Table 9

performance improvements were achieved by the proposed framework over the competing approaches on the daily brent cop data.

Method	[21]	[57]	[58]	RF	ARF	[30]	[59]	[60]	[16]	[61]	[62]	[28]	[59]
MAPE	83.540	96.963	85.255	94.816	87.713	87.770	62.201	54.922	70.582	92.172	92.528	68.882	57.789
MSE	94.637	99.864	95.691	99.514	98.746	96.708	72.162	56.063	78.846	98.734	99.001	79.758	60.957
RMSE	76.842	96.310	79.241	93.026	88.802	81.856	47.238	33.716	54.006	88.750	90.003	55.009	37.516
MAE	85.153	95.420	86.338	92.942	81.453	86.994	66.250	59.782	73.236	93.331	91.967	72.170	62.048
R2	0.943	58.213	1.176	11.480	11.304	1.545	0.180	0.114	0.237	4.120	5.276	0.248	0.128
EVS	0.930	51.567	1.163	10.561	10.427	1.465	0.167	0.102	0.150	3.929	4.380	0.235	0.116

Table 10

Performance improvements were achieved by the proposed framework over the competing approaches on the weekly brent cop data.

Method	[21]	[57]	[58]	RF	ARF	[30]	[59]	[60]	[16]	[61]	[62]	[28]	[59]
MAPE	68.797	90.029	73.884	84.389	80.794	77.735	43.342	14.404	26.540	77.292	72.514	67.407	68.797
MSE	90.539	99.200	94.017	97.656	96.351	94.773	68.373	23.736	44.342	95.074	94.299	89.575	90.539
RMSE	69.242	91.057	75.541	84.691	80.898	77.138	43.762	12.671	25.396	77.806	76.122	67.712	69.242
MAE	70.779	86.357	74.063	80.556	74.021	76.516	45.378	21.538	34.739	79.626	74.710	68.895	70.779
R2	3.057	55.851	4.966	13.876	13.900	5.737	0.846	0.308	0.449	6.114	5.228	2.760	3.057
EVS	3.113	48.828	5.021	12.844	12.819	5.690	0.895	0.362	0.458	6.037	5.012	2.815	3.113

Table 11

Findings obtained from the diebold–mariano test for comparing the forecasts from the proposed against competing methods on brent cop data.

Pair comparison			Daily			Weekly		
Target	Competing		MAPE	MAE	RMSE	MAPE	MAE	RMSE
Proposed	LR [21]	DM	11.389	5.540	4.159	8.070	2.779	4.486
		P-value	0.049	0.046	0.025	0.030	0.032	0.022
	SVR [57]	DM	9.134	16.313	22.630	−4.628	14.224	21.279
		P-value	0.024	0.009	0.024	0.019	0.021	0.019
	ARF	DM	4.621	11.212	−3.809	10.011	18.214	17.342
		P-value	0.009	0.022	0.0014	0.0132	0.009	0.039
	DNN [58]	DM	32.523	5.110	4.130	6.838	9.608	−12.267
		P-value	0.007	0.015	0.019	0.026	0.015	0.020
	LSTM [30]	DM	14.040	17.507	6.775	−9.539	3.945	3.750
		P-value	0.013	0.028	0.019	0.042	0.036	0.047
	CEEMDAN RF[59]	DM	−7.368	8.782	14.073	8.185	12.708	−4.144
		P-value	0.019	0.049	0.033	0.008	0.023	0.012
	CEEMDAN LSTM [60]	DM	23.607	16.044	−7.251	15.675	21.829	27.894
		P-value	0.017	0.027	0.037	0.015	0.017	0.019
	CEEMDAN LSTM_CNN [16]	DM	29.041	10.500	−13.745	5.173	8.181	−31.361
		P-value	0.039	0.044	0.046	0.003	0.048	0.016
	EWT_BILSTM [61]	DM	5.150	−4.079	16.027	3.943	−24.607	31.880
		P-value	0.045	0.036	0.028	0.041	0.005	0.006
	VMD_LSTM_ATT_SVR [62]	DM	5.379	−1.990	6.168	21.111	32.009	26.944
		P-value	0.032	0.021	0.044	0.037	0.047	0.001
VMD_CONV_GRU [28]	DM	10.101	11.022	18.348	2.485	17.812	−2.368	
	P-value	0.016	0.033	0.024	0.010	0.043	0.017	
CEEMDAN RF_LSTM [59]	DM	1.999	−20.684	16.123	−5.690	16.686	9.258	
	P-value	0.018	0.044	0.003	0.012	0.029	0.020	

findings reveal that the proposed framework has a competing predictive modeling capability that enables it to overcome the cutting-edge methods for COP prediction.

6.2. Statistical analysis

In order to validate the predictive efficiency of the proposed framework against the competing methods, we need to determine if there are statistically significant differences among their results. To achieve that, a Diebold–Mariano (DM) test (Diebold, 1995), which set the null hypothesis as “there is no distinction among the two forecasting errors”, is applied to compare the assess the statistical significance of our results against those obtained from each competing methods. Again, the null hypothesis is rejected when the computed P-value is less than 0.05 and infer that there is a substantial variation in predictions from the two models. Table 11 shows the results of the DM test for pair comparisons between the proposed framework and competing methods according to RMSE, MAE, and MAPE of forecasting performance on both daily and weekly Brent COP data. The majority of DM findings are significant at the 0.05 level of p-values. Consequently, data from the proposed framework and competing models provide statistically substantial evidence that they are distinct.

6.3. Ablation analysis

This section provides experimental ablation analysis to further study and understand the contribution of different building blocks to final predictive modeling performance.

6.3.1. Hyperparameter optimizers

To analyze and demonstrated the impact of TPE in optimizing the hyperparameters of the proposed framework, we compare the prediction performance of the proposed framework, when its hyperparameters are calculated by TPE, grid search, a random search for 30 trials on Brent COP data. The comparative results of these experiments are illustrated in Fig. 8 for daily and weekly COP data. In section 2, we noticed that PSO and ABC are common algorithms for optimizing the hyperparameters of predictive models. So, the performance of extend TPE in our framework is also compared against PSO and ABC. it is obvious that the optimum hyperparameters settings discovered by TPE show better performance than those obtained by random and grid search in the deemed daily and weekly case studies. This can be attributed to the fact that random search generates a high variance during computation as the hyperparameters are selected at random with no intelligence. Besides, grid search suffers from dimensionality problem, with complexity of $O(n^k)$, and did not guarantee reaching the ideal solution especially with large

number of hyperparameters k . More interestingly, the presented TPE also overcome the PSO and ABC across different performance measures validating the efficiency of the hybrid optimization of decomposition as well as learners' hyperparameters. when dealing with PSO and ABC optimizers, it is challenging to regress the behavior according to the separate rules, and the functions of a colony could not be understood without identifying in what way an agent workings. One more issue is that even negligible variations in the modest rules reason changes in behavior of swarm. From computation point of view, TPE and PSO are sharing a computational complexity of $O(n \log n)$, (Yang & Shami, 2020) making achieved improvements considerable.

6.3.2. Dropout regularization

This section compares the proposed framework against one that uses the same design and yet is trained without dropout. The remaining hyperparameters and configurations of the framework are kept consistent. TPE optimizes the likelihood of the employed dropout dynamically. Both the daily and weekly Brent data are used to conduct a complete analysis of dropout throughout the testing stages, as shown in Fig. 9. The comparison of the findings indicates that the proposed framework trained without dropout had fewer training errors but significantly larger test errors, which may be an indication of overfitting. In comparison, the dropout model considerably decreases the overfitting trouble across all datasets, resulting in decreased test errors. The average error decline achieved by the dropout model is percent, indicating the usefulness of dropout in minimizing overfitting and boosting the predictive modeling capability of the DL model.

6.3.3. Decomposition methods

To understand and analyze the effectiveness of OVDM, the proposed framework, multiple experiments are performed by evaluating the proposed framework when trained with different decomposition techniques namely EMD (Huang et al., 1998), EEMD (Wu & Huang, 2009), CEEMD (Yeh, Shieh, & Huang, 2010), and VMD (Konstantin & Dominique, 2014). The VMD parameters is statically set as presented in (Konstantin & Dominique, 2014). In each experiment, all layers and hyper-parameters of the proposed framework are kept identical to assure fair analysis. The prediction performance obtained from these experiments is displayed and compared for both daily and weekly Brent COP data, as shown in Fig. 10. It is worth noting that EMD and EEMD achieve the worst prediction performance on both daily and weekly data across different metrics. Also, it could be noted that OVMD achieves a robust predictive modeling performance overcoming its competing techniques including standard VMD. This can be attributed to its ability to optimize its hyper-parameters using TPE algorithm according to the prediction loss function, which lead to improve the predictive modeling

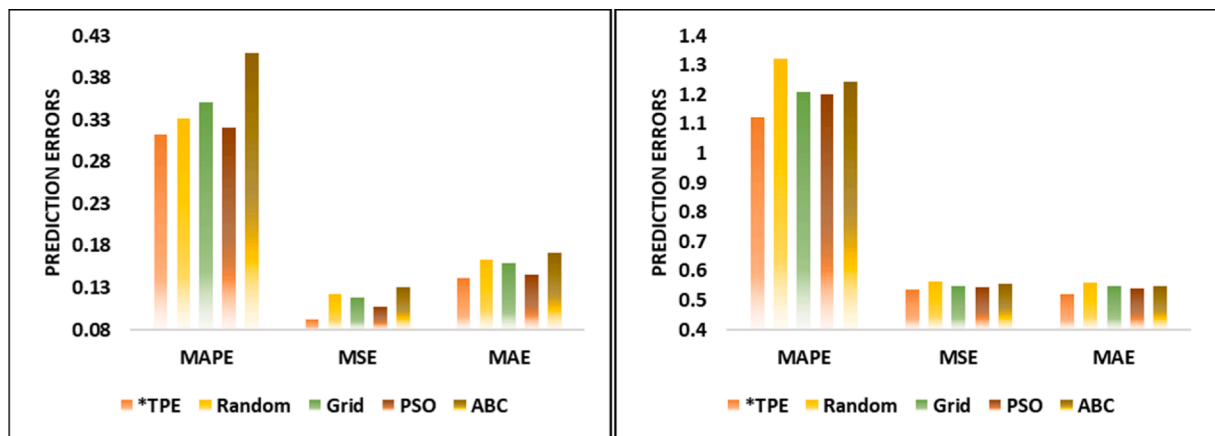


Fig. 8. Predictive performance of the proposed framework under different hyper-parameter optimization methods. A) Results on daily Brent COP data. B) Results on weekly Brent COP data.

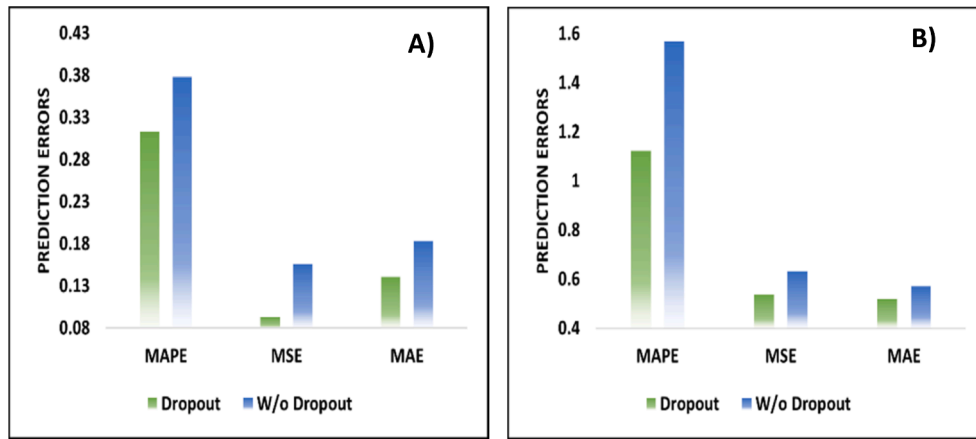


Fig. 9. Comparison of the predictive performance of the proposed framework when trained with and without Dropout. A) Results on daily Brent COP data. B) Results on daily weekly COP data.

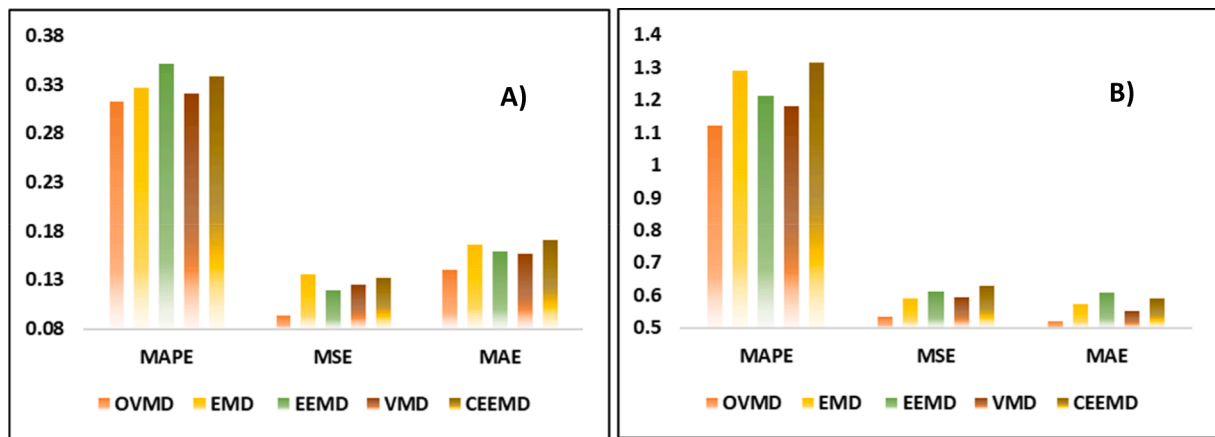


Fig. 10. Comparison of the predictive performance of the proposed framework when trained with different decomposition techniques. A) Results on daily Brent COP data. B) Results on daily weekly COP data.

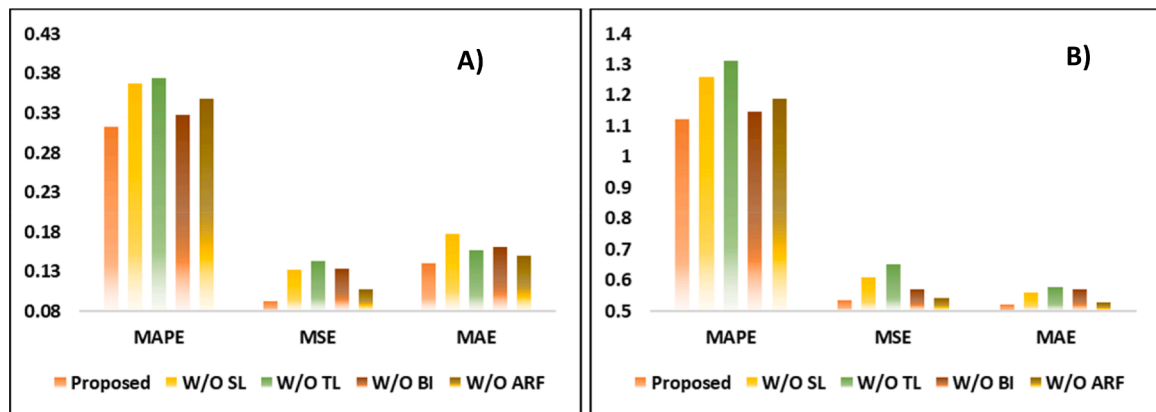


Fig. 11. Comparison of the predictive performance of the proposed framework when eliminating different building blocks. A) Results on daily Brent COP data. B) Results on daily weekly COP data.

capability.

6.3.4. Learning modules

To validate the efficacy of different building blocks in the proposed framework, comparative ablation experiments are performed to compare the predictive modeling performance of the proposed framework when implemented without particular learning modules. The

results of these experiments are visualized for different metrics in Fig. 11 for both daily and weekly data. It is notable that the removal of SL module or TL modules leads to a significant increase in prediction errors, which implies their fundamental roles in improving the predictive modeling capabilities. On the other hand, it could be seen that the removal of the bidirectional learning module negatively influences the predictive modeling performance on both daily and weekly data. This in

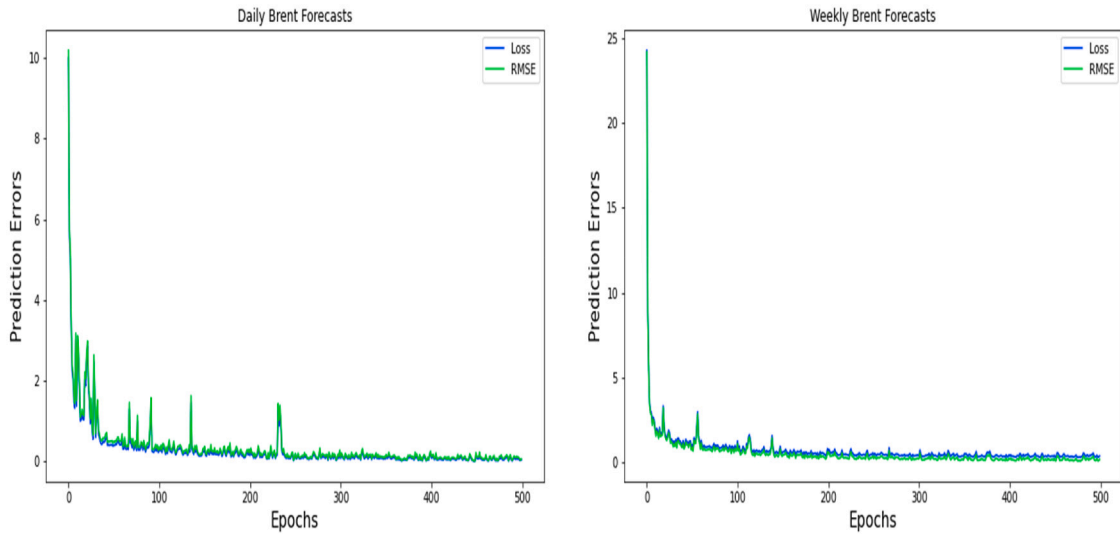


Fig. 12. The training curves of the proposed framework on daily and weekly Brent dataset.

turn validates the hypothesis that backward COP information is essential for efficient forecasting. It is also worth mentioning that the ARF algorithm considerably contributes to improving the predictive capabilities on daily (MAPE: 0.349, MSE: 0.108, MAE: 0.15) and weekly data (MAPE: 1.189, MSE: 0.544, MAE: 0.529). This can be attributed to its ability to model high variations and changes in the high frequency time series.

6.4. Convergence analysis

This section presented the training and validation loss curves for both the daily and weekly Brent COP datasets in Fig. 12 to interpret and analyze the training performance of the proposed solution. For the daily COP data, the framework displays steady learning behavior over training epochs without any under-fitting and/or overfitting or concerns, suggesting smooth gradient computations. After a hundred training epochs, it begins to converge while still providing robust validation performance during the training process itself. In contrast, for the weekly COP data, the proposed framework exhibits fast convergence after 50 epochs on the weekly COP data, while retaining clean training and validation curves, demonstrating consistent learning. There's also a substantial correlation between training and validation results throughout training. Moreover, Fig. 13 and Fig. 14 depicts the compare the predictions obtained from the proposed framework with

corresponding actual values in the unseen test set for both daily and weekly data, correspondingly. The plotting demonstrates how closely the framework predictions match up with actual values, which further supports our findings in the previous sections.

6.5. Complexity analysis

This section further investigates the performance of the proposed framework from the time efficiency perspective. As shown in Table 9 and Table 10, the execution time of the proposed framework is compared against the competing methods and the results show that the proposed framework succeeds to achieve robust predictive performance while maintaining acceptable time efficiency. This in turn validates the suitability of the proposed solutions to be applied and integrated in real-world oil markets. Apart from the numerical time analysis of the overall framework, we provide an asymptotic analysis of different building blocks of the proposed framework. The analysis of computational complexity of VMD relies on the recursive rapid Fourier transform and initialization of the base frequency of every mode and the (Karasu & Altan, 2022). The computing stages primarily involve 1) base frequency initialization with the complexity of $O(2N \log(2N))$; 2) iterative mode update $\tilde{u}_{n+1}^i$ with the complexity of $O((6A + 2M + 2D) \bullet I \bullet MI \bullet 2N)$; 3)

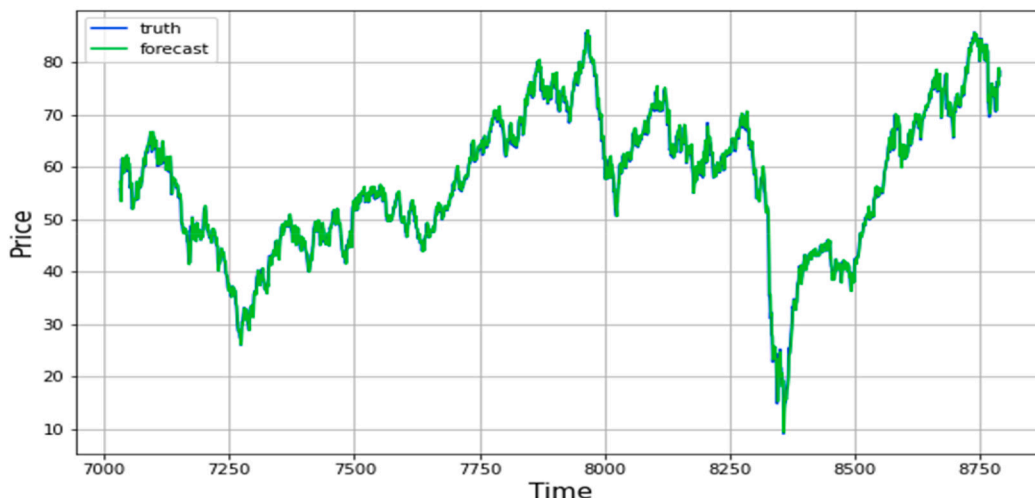


Fig. 13. Visual comparison between the forecasts generated by the proposed framework and the actual (truth) value in the test set of daily COP data.

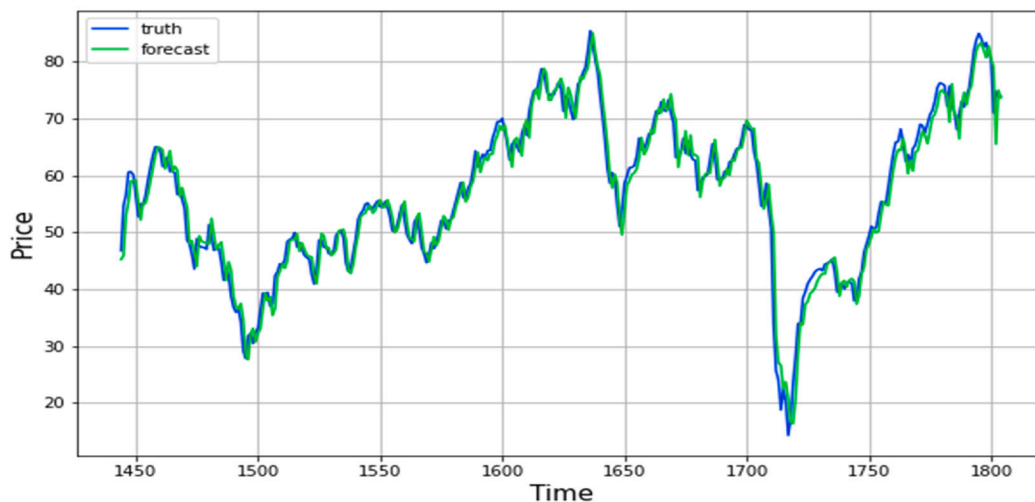


Fig. 14. Visual comparison between the forecasts generated by the proposed framework and the actual (truth) value in the test set of daily COP data.

iterative frequency update $\tilde{w}_{n+1}^i$ with the complexity of $O((2C + 3M + 2A) \bullet I \bullet MI \bullet N)$; 4) the convergence operation with the complexity of $O((4A + 2M) \bullet MI \bullet 2N)$, and 5) the data construction process has a complexity of $O(2N \log(2N))$. Whereas A, C, D, M denote the addition, comparison, division, and multiplication, correspondingly. Moreover, N, I, and MI denote the length of the COP data to be decomposed, the number of modes to be decomposed, and the highest number of update iterations. This way, the final computational complexity of the VMD of $O(2N \log(2N))$. Regarding the spatial learning module, given that k as kernel size of convolutional layer, I as the number of input channel of convolutional layer, Y as the number output channels of convolutional layers, and L as the input length, then the number of convolutional filters can be computed to be $(L \bullet K \bullet I)$. Next, the convolutional filters are mapped to a different number of channels along with the bias. Thus, the computational complexity of the convolutional can be asymptotically defined as $O(((L \bullet K \bullet I) + 1) \bullet Y)$.

7. Conclusions and future work

This work presents a novel predictive modeling framework that can efficiently learn the spatial-temporal representations necessary to forecast the future crude oil prices at the daily and weekly resolutions. A novel optimized VMD is introduced to decompose the Brent COP data to multiple modes in such a way that maximize the predictive modeling performance. ARF is introduced to perform predictive modeling on high-frequency modes. Also, the representational power of the proposed framework is empowered by combining spatial, temporal, and bi-directional representation learning capabilities in a single deep network. Contrasted with the competing solutions, the proposed framework shows many advantages: 1) Through OVMD decomposition, the oscillation features, and core regulations of COP series at diverse resolutions are profoundly explored; 2) The new predictive deep learning model enables dealing with the challenge of learning from the dynamic and non-linear COP time series without affecting the performance of the model. 3) it can improve the predictive representation capabilities without incurring additional computing costs. By applying this high-accuracy solution, the dilemma of COP is profoundly analyzed, the future variability condition of COP price is subdued, and the countermeasures against COP risks will be articulated appropriately, to reach sustainable risk management for COP in both local and global oil markets. Experimental evaluations on Brent oil data validate the superiority of deep learning over the competing methods across different performance metrics such as efficiency RMSE, MAE, MAPE under daily and weekly forecasting scenarios.

Finally, we present four promising directions that can be investigated

to extend this work in the future. First, an important extension of this work resides in collecting benchmark data that includes the considerations for improving the predictivity of COP in the real world. These considerations are not restricted to environmental factors but include commercial and political factors across the world. For example, the war between Russia and Ukraine shows that many risks to global COP and food security have been raised. To better cope with these changes, an adaptive predictive modeling solution is of great importance to empowering risk management and decision-making in such dynamic scenarios. Deep reinforcement learning (K. Li, Zhang, & Wang, 2020) and online learning (Hoi, Sahoo, Lu, & Zhao, 2021) are promising learning schemas to be investigated in this regard. Second, to better support the sustainable management of future global and local stock markets, this work can be improved in the future by studying predictive modeling of COP in conjunction with gold prices, metal prices, stock prices, and prices of other energy sources. This in turn will help realize the concept of smart risk management, especially in developing countries that are lacking such resources. Collaborative learning over the internet can show great promise in this regard. Third, some risk management and decision-making tasks may require predicting COP for multiple days or multiple weeks in the future, which may entail high uncertainties. Therefore, this study can be extended in the future to learn to generate probabilistic multi-step predictive modeling, whereas the output at each time-step can be supported with an uncertainty ratio and/or confidence value that can guide the decision makers about the quality of the generated forecasts. Bayesian learning (H. Wang & Yeung, 2020) and fuzzy learning systems (de Lima Silva, Sadaei, Ballini, & Guimarães, 2019) are promising methods in this regard. Fourth, as is the same case with most deep learning models, the proposed framework is considered a black-box solution as it is hard for managers, stack holders of oil markets, and other practitioners to understand and interpret its internal logic and the decisions it generates. Thus, in future work, we can extend this work to design an explainable predictive modeling solution by designing a new model-agnostic explanation method to generate some counterfactual explanation for the obtained forecasts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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